

Dr. Vishwanath Karad

**MIT WORLD PEACE
UNIVERSITY | PUNE**

TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

PROGRAM STRUCTURE AND SYLLABUS

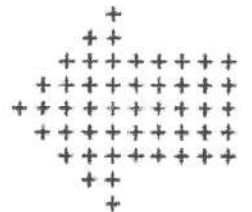
FACULTY OF ENGINEERING AND TECHNOLOGY

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

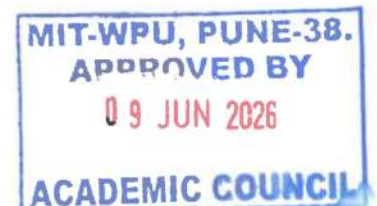
DEPARTMENT OF COMPUTER SCIENCE AND APPLICATIONS

Master of Computer Applications

BATCH 2026 - 2028



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Preamble:

First year provides foundation of basic computer fundamental courses along with spiritual sessions. This is semester pattern and first year builds foundation for entire program which will provide complete exposure of prerequisites which are necessary for advance learning from second year onwards. Spiritual sessions will help the students grooming from social and ethical behaviour point of view.

Second year courses are designed with specialization approach which would help students to develop basic and advanced skills in areas of their interest thereby increasing their level of expertise. Mini Project during the third semester will be based on open-source technologies learned in first year.

At the end of the second-year students will be exposed to student's internship/exchange programs based on the specializations. Second year will be focused on Industry sponsored Research project on Innovation in Technology belonging to the specialization approach for six months. Industry Internship programs for the students which will be as good as pre-placement exposure with the industry to work on live projects. Industry Internship programs helps the students to learn managerial and leadership skills, problem solving skills, team spirit, and professional ethics.

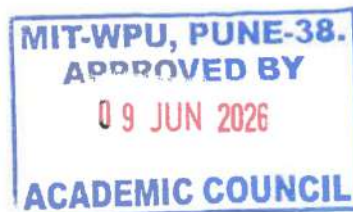
Intended philosophy of the syllabus is to meet following guidelines:

Give strong foundation on core Computer Science and application courses. Expose student to emerging trends in a gradual and incremental way.

Offer specialization on a chosen area.

Create research temper among students in the whole process.

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Vision and Mission of the Department:

VISION

To contribute to society through excellence in scientific and knowledge-based education utilizing the potential of computer science with a deep passion for wisdom, culture, and values.

MISSION

- To create knowledge, to disseminate knowledge, and to provide service to our society.
 - Provide quality undergraduate and graduate education in both the theoretical and applied foundations of computer science.
 - Train students to effectively apply this education to solve real-world problems thus amplifying their potential for lifelong high-quality careers.
 - To give them a competitive advantage in the ever-changing and challenging global work environment
 - To achieve a distinguished position in Computer Science through innovative teaching, learning methods and research.
- To develop strong fundamentals and habit of life-long learning in students to fulfil the needs of industry.

Programme Educational Objectives:

Computer Application Postgraduates will have following accomplishments within a few years of graduation:

PEO1	To Analyse the real-world problems, design computer applications that are acceptable by the industry.
PEO2	To have professionalism, ethical attitude, communication skills, ability to work in team and adapt recent trends in the technology through lifelong learning.

Programme Outcomes:

PO1	Students will be able to apply fundamental knowledge of Computing, Mathematics, Science, and Engineering appropriate to the discipline.
PO2	Students will be able to analyse a problem, identify and formulate the computing solution(s) with appropriate model.
PO3	Students will be able to analyse the local and global impact of computing on individuals, organizations, and society.
PO4	Students can understand the professional, ethical, legal, security and social issues and responsibilities.

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PO5	Students can achieve continuous professional development through long- term learning.
PO6	Students can understand engineering and management principles and apply these to one's own work, as a member and a leader in a team, to manage projects.

Programme Specific Outcomes (PSOs):

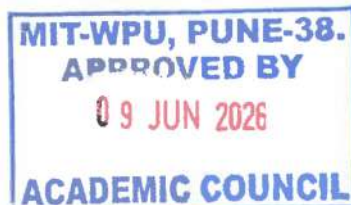
PSO-1	To understand and apply the concepts of computer applications for industrial real time problem solving.
PSO-2	To analyse, design, develop, test, and maintain the software applications with latest technologies and computing tools.

Programme Structure:

- Programme Duration:** Two Years
- System followed :** Semester
- Credits System (As per Academic Ordinance 2023-24):** A unit by which the course work assigned to the student is measured. It determines the number of hours of instructions required per week. One credit is equivalent to one hour of supervised learning (SL) and two hours of lab session per week. Considering the self-learning requirement, peer learning and faculty facilitation, the student shall have maximum 3 or 4 hours of supervised learning and 2 hours of lab session time. One Credit defined as 15 hrs for theory session and 30 hrs of lab session for 1 credit. For internship/project work or field work one credit defined as 3 hrs of industrial internship/project work. Teaching Learning Process at MITWPU is designed to ensure this. Student will be typically taking 18-23 Credits per Semester of 15 weeks. A 2-year course will mean completing 84 credits.
- Credit distribution for PG Programs:**

	% Credit Allotment	Credit Assigned
Program Major (PM)	48.81	41
Program electives (PE)	19.05	16
Program Capstone Project, Problem Based Learning, Seminar, and Internships (PR)	21.43	18
University core (UC)	10.71	9
Total	100	84

Program Major, Program Foundation and Program Capstone Project, Problem Based Learning, Seminar and Internships have been clubbed together as Program Core (PC) in the structure.



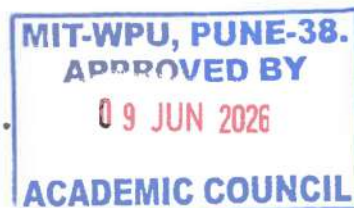
MITWPU Course codes with effect from AY 2024-25

Course Code Logic

1. The course code is of the format **AAAXYYYYZ**. It has total eight characters comprising alphabets/digits.
2. **AAA**: The first characters (alphabets) stand for an exclusive short code of the Program owning the course (Such as PHY for Physics, CHM for Chemistry, MST for Maths and Statistics, MSE for Materials Science and Engineering, COS for Computer Science, CSE for Computer Science and Engineering, CHE for Chemical Engineering etc.).
3. **X**: Fourth character (numeric) stands for course level
 - a. Level 0 for bridge level/introductory-level diploma courses
 - b. Level 1 stands for core diploma/foundation/introductory level UG courses
 - c. Level 2 stands for core UG courses
 - d. Level 3 stands for advanced UG courses
 - e. Level 4 stands for specialization courses (generally final year UG and first year PG courses and PhD courses)
 - f. Level 5 stands for specialization PG courses and PhD courses
4. **YYY**: Fifth, sixth and seventh characters (numerics) stand for the serial number of the course given by the respective Program.
5. **Z**: Eighth character (numeric) stands for the version of the same course. For example, '0' is the first version, '1' is second version and so on. There will be change of version in course if the contents are modified (more than 30%) after one academic year by the BoS.
6. Note for University Core Courses: University core courses codes will be assigned centrally by the office of Dean Academics.

Note for University Electives: The course code for the University Elective will be assigned by the Program offering the University Elective.

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Assessment Scheme

Abbreviation	L-T-P-J-C : L-Lecture, T-Tutorial, P-Practical, J-Project, C-Total Credits. CCA1 - Class Continuous Assessment 1, MT - Mid Term Test, CCA2 - Class Continuous Assessment 2, LCA1 - Laboratory Continuous Assessment 1, LCA2 - Laboratory Continuous Assessment 2, LCA3 - Laboratory Continuous Assessment 3, TE - Term End Exam
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Sr. No.	Type of Course	Assessment Scheme Code	Description L-T-P-J-C	CCA1	MT	CCA2	LCA1	LCA2	LCA3	TE	Total
1	Theory Courses	TT1	All Theory (L, T) Only courses with TE exams	15	30	15	-	-	-	40	100
2	Theory Courses with Continuous Evaluation	TT2	All Theory (L, T) only courses without TE exams	35	30	35	-	-	-	-	100
3	Lab /Projects/ Internship/ Dissertation	PJ	All courses having P and J components Only	-	-	-	33	33	34	-	100
4	Theory and Lab Course 1	TL1	2-0-2-0-4	7	15	8	10	10	10	40*	100
5	Theory and Lab Course 2	TL2	1-0-3-0-4	2	10	3	15	15	15	40*	100
6	Theory and Lab Course 3	TL3	3-0-1-0-4	10	25	10	5	5	5	40	100
7	Theory and Lab Course 4	TL4	2-0-1-0-3	10	20	10	6	7	7	40	100
8	Theory and Lab Course 5	TL5	1-0-2-0-3	5	10	5	13	13	14	40*	100
9	Theory and Lab Course 6	TL6	2-1-1-0-4	10	25	10	5	5	5	40	100
10	Theory and Lab Course 7	TL7	1-1-1-0-3	10	20	10	6	7	7	40	100
11	Theory and Lab Course 8	TL8	4-0-1-0-5	11	25	12	4	4	4	40	100
12	Theory and Lab Course 9	TL9	3-0-2-0-5	9	24	9	6	6	6	40	100

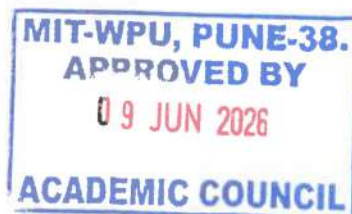
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13	Theory and Lab Course 10	TL10	2-0-3-0-5	7	10	7	12	12	12	40	100
14	Theory and Lab Course 11	TL11	1-0-4-0-5	2	7	3	16	16	16	40	100
15	Theory and Lab Course 12	TL12	1-0-1-0-2	7	15	8	10	10	10	40*	100
16	Theory and project Course without term end	PJ	Any Combination of lab and theory credits	--	--	--	33	33	34	--	100
17	MOOC Courses	M**	Pre-approved MOOC courses with number of credits as given in the syllabus	--	--	--	--	--	--	--	100
18	Immersive Courses such as Cocreation, Rural Immersion, NAIP	I***	Credits as per syllabus	--	--	--	--	--	--	--	100
19	Lab /Projects/ Internship/ Dissertation with End Term (End-term Lab examination)	PR	Credits as per syllabus	--	--	--	30	30	--	40	100
20	Jury Based Evaluation (End-term Jury examination)	J	Credits as per syllabus	30	--	30	--	--	--	40	100
21	Mandatory Learning Course	MLC	Credits as per syllabus	Complete / Non - Complete							
Peace Courses (University Core)											
Sr. No.	Type of Course	Assessment Scheme Code	Description L-T-P-J-C	CCA1	CCA2	CCA3	LCA1	LCA2	LCA3	TE	Total
22	Peace Courses (University Core)	UP	Peace Courses (University Core)	15	15	30	--	--	--	40	100

* Term End Exams to be conducted anywhere within the MITWPU Campus subject to the following conditions:

23/6/2026



1. All eligible students will be taking this exam in the same space and at the same time slot.
2. The time for Term End Exams will be a maximum of 3 hrs.
3. QP will be sent along with Invigilators by DoE.

** Assessment of MOOC courses including the conversion of the grade / score / percentage obtained therein will be as per the MOOC policy approved by the BOE.

*** Assessment of Immersive Courses such as Cocreation, Rural Immersion, NAIP will be done as per pre-approved rubrics.



MCA (First Year) (Batch 2026-28)

Semester – I

Sr. No.	Course Code	Name of the Course	Type	Weekly Workload, Hrs.				Credits	Assessment Scheme Code
				L	T	P	J		
1	MCA40140	Data Communication and Security	PM	4				4	TT1
2	MCA40150	Java Programming	PM	3				3	TT1
3	MCA40160	Advanced Database Management System	PM	3				3	TT1
4	MCA40170	Python Programming	PM	3	1			4	TT1
5	MCA40180	Lab on Java Programming	PM			2		1	PR
6	MCA40190	Lab on Advanced Database Management System	PM			2		1	PR
7	DCA40010	Research Methodology	UC	4				4	TT1
8	PCE10040	Scientific Studies of Mind, Matter, Spirit, and Consciousness	UC	2				2	UP
9	YOG10030	Yoga	UC			2		1	PJ
Total				19	1	6	--	23	--
Weekly Total Teaching Hours				26				--	--

L-Lecture, T-Tutorial, P-Practical, J-Project

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MCA (First Year) (Batch 2026-28)

Semester – II

Sr. No.	Course Code	Name of the Course	Type	Weekly Workload, Hrs.				Credits	Assessment Scheme Code
				L	T	P	J		
1	MCA40200	Advanced Java	PM	3				3	TT1
2	MCA40210	Data Structures and Algorithms	PM	3				3	TT1
3	MCA40220	Statistics and Probability	PM	4				4	TT1
4	MCA40230	Lab on Advanced Java	PM			2		1	PR
5	MCA40240	Lab on Data Structures and Algorithms	PM			2		1	PR
6		Program Elective – I (PE1)	PE	3	1			4	TT1
7		Program Elective – II (PE2)	PE	3	1			4	TT1
8	DCA40020	Cognitive Skills	PR			2		1	PJ
9	PCE10050	Peace Building: Global Initiatives	UC	2				2	UP
Total				18	2	6	--	23	--
Weekly Total Teaching Hours				. 26				--	--

L-Lecture, T-Tutorial, P-Practical, J-Project

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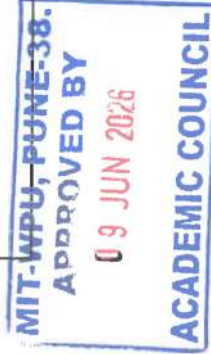
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MCA (Second Year) (Batch 2026-28)

Semester – III

Sr. No.	Course Code	Name of the Course	Type	Weekly Workload, Hrs.				Credits	Assessment Scheme Code
				L	T	P	J		
1	MCA40260	Devops	PM	3				3	TT1
2	MCA40270	Full Stack Development	PM	3				3	TT1
3	MCA40280	Software Testing and Quality Assurance	PM	3				3	TT1
4	MCA40290	Lab on Full Stack Development	PM			2		1	PR
5		Program Elective – III (PE3)	PE	3	1			4	TT1
6		Program Elective – IV (PE4)	PE	3	1			4	TT1
7	MCA40250	Research Paper Writing	PR		1			1	PJ
8	MCA40300	Mini Project	PR				3	1	PJ
Total				15	3	2	3	20	--
Weekly Total Teaching Hours				23				--	--

L-Lecture, T-Tutorial, P-Practical, J-Project









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MCA (Second Year) (Batch 2020-28)

Semester – IV

Sr. No.	Course Code	Name of the Course	Type	Weekly Workload, Hrs				Credits	Assessment Scheme Code
				L	T	P	J		
1		MOOC	PM	3				3	M
2	DCA50010	Internship/Capstone Project	PR				45	15	PR
Total				3	--	--	45	18	--
Weekly Total Teaching Hours				48				--	--

L-Lecture, T-Tutorial, P-Practical, J-Project

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DoCSA, SoCSE

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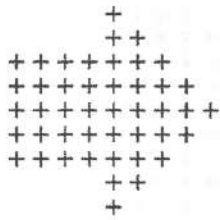
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Program Elective	Semester	Course Code	Name of the Course
Program Elective-I	II	MCA50170	Data Science and Analytics
Program Elective-I	II	MCA50180	Mobile Application Development - iOS Programming
Program Elective-I	II	MCA50190	Next Generation Database
Program Elective-I	II	MCA50200	Cloud Computing
Program Elective-II	II	MCA50210	Artificial Intelligence and Machine Learning
Program Elective-II	II	MCA50220	Advanced Software Development Methodologies
Program Elective-II	II	MCA50230	Data mining and Data Warehousing
Program Elective-II	II	MCA50240	Internet of Things
Program Elective-III	III	MCA50250	Deep Learning and Introduction to NLP
Program Elective-III	III	MCA50260	Advanced Internet Technology
Program Elective-III	III	MCA50270	Data Engineering
Program Elective-III	III	MCA50280	Information Security Audit
Program Elective-IV	III	MCA50290	Computer Vision & Digital Image Processing
Program Elective-IV	III	MCA50300	ReactJS
Program Elective-IV	III	MCA50310	Big Data Analytics
Program Elective-IV	III	MCA50320	Data Privacy and Security

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PROGRAM STRUCTURE AND SYLLABUS

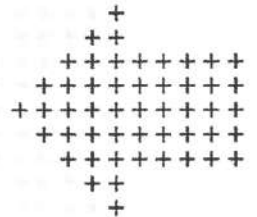
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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE AND APPLICATIONS

Master of Computer Applications

BATCH 2026 - 2028



			
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Dr. Maheshwari Kulkarni
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COURSE STRUCTURE

Course Code	MCA40140				
Course Title	Data Communication and Security				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	4				4
Credits	4				4
Assessment Schema Code	TT1				

Prerequisites:

Basic knowledge of computer fundamentals.

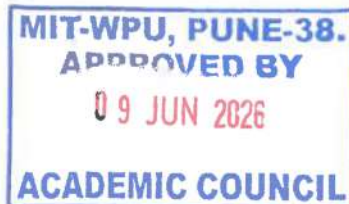
Course Objectives:

1. To provide a comprehensive understanding of the architecture, protocols, and functionalities of computer networks, including the OSI and TCP/IP reference models.
2. To introduce key concepts in data communication, such as transmission media, encoding, modulation, and switching techniques.
3. To explain the design and operation of networking layers, with an emphasis on addressing, routing, congestion control, and transport protocols.
4. To develop an understanding of application layer protocols and their role in enabling network-based services.
5. To familiarize students with basic concepts in network security, including common threats and fundamental techniques for protecting data and communication systems.

Course Outcomes:

After completion of this course students will be able to:

1. Describe and explain the structure and operation of computer networks, including the OSI and TCP/IP models, and the functionality of each layer.
2. Analyse and apply techniques for error detection and correction, framing, and channel allocation to enhance data transmission reliability.
3. Analyse network addressing schemes, routing protocols, and congestion control techniques for efficient and effective data delivery.
4. Interpret the roles and functionalities of transport and application layer protocols, such as TCP, UDP, DNS, SMTP, and FTP, in network communication.



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5. Identify common network security threats and explain foundational security measures, including cryptographic concepts, firewalls, and VPNs, to protect data and communication systems.

Course Contents:

Unit 1: Introduction and Physical Layer (15 Sessions)

Overview of Computer Network, OSI and TCP/IP Reference Models, Guided and Unguided Transmission Media, Analog and Digital Communication, Encoding and Modulation, Nyquist Theorem, Shannon's capacity, Switching techniques- TDM, FDM.

Unit 2: Data Link Layer (15 Sessions)

Framing, Error detection and Error correction, Vertical Redundancy Check (VRC), Longitudinal Redundancy Check (LRC), Hamming Distance, Hamming Code, Cyclic Redundancy Check (CRC), Stop and Wait Protocol, Sliding Window Protocol, Go-Back-n ARQ, Selective-Retry ARQ, HDLC, Channel Allocation, ALOHA Systems, CSMA Protocols, Collision Free Protocols

Unit 3: Network Layer (10 sessions)

Introduction to the Network Layer, Network Layer Protocols IPv4 and IPv6: header structure and features, Comparison of IPv4 and IPv6, Transition strategies (Dual stack, Tunneling), Addressing in the Network Layer, Classful vs Classless addressing, Routing Fundamentals, Types of routing: static, dynamic, Routing protocol, Causes and effects of network congestion, Congestion control techniques: Leaky Bucket and Token Bucket, Network Address Translation (NAT)

Unit 4: Transport Layer (5 sessions)

Process-To-Process delivery, user datagram, Transmission control protocol.

Unit 5: Application Layer (5 sessions)

Client-Server Model, Socket interface, A brief introduction to DNS, SMTP, FTP.

Unit-6: Network Security (10 sessions)

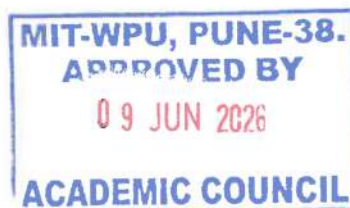
Introduction to Network Security, Common threats: malware, phishing, and DoS attacks, Cryptography Basics, plaintext, ciphertext, encryption, decryption, Authentication and Authorization, passwords, two-factor authentication, and biometrics, Firewalls and VPNs

Learning Resources:

Reference Books:

1. Data Communications & Networking, Behrouz A. Forouzan, Tata McGrawHill, 4th Edition,

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Blade.



- 2007
2. Data Communication and Computer Networks, Rajneesh Agarwal, Vikas Publication house, 2018, reprint
 3. Computer Networks by Andrew Tanenbaum, Fourth Impression, Pearson Education India, 2010
 4. Internetworking with TCP/IP, Vol. 3, Client-Server Programming and Applications by Douglas E. Comer, Prentice Hall Publisher, 2014

Supplementary Reading:

Web Resources:

1. <https://www.javatpoint.com/computer-network-tcp-ip-model>
2. <https://www.geeksforgeeks.org/classes-of-routing-protocols>

MOOCs: Online courses for self-learning

Courses by NPTEL and MIT Open Courseware etc.
<https://nptel.ac.in/courses/106105082>

Pedagogy:

- Participative learning
- Discussions
- Problem solving
- Power point presentations
- Videos
- Demonstrations

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COURSE STRUCTURE

Course Code	MCA40150				
Course Title	Java Programming				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3				3
Credits	3				3
Assessment Schema Code	TT1				
<u>Prerequisites:</u>					
Basic programming knowledge of C and C++, concept of object-oriented programming.					
<u>Course Objectives:</u>					
1. To enable the students to understand the core principles of the Java Language and use visual tools to produce well designed, effective applications and applets. 2. To make students understand and learn the basic Java programming. 3. To make students build desktop-based applications using Java					
<u>Course Outcomes:</u>					
After completion of this course students will be able to: 1. Create Java application using object-oriented concepts. 2. Understand and apply the concepts of exception handling. 3. Apply the concepts of multithreading, java collections and file handling.					
<u>Course Contents:</u>					
Unit 1: Fundamentals of Java (4 Sessions)					
Object, classes, abstraction, encapsulation, inheritance, polymorphism, Identifiers and Keywords Data Types in Java coding Conventions, Expressions in Java Control structures, decision making statements Arrays and its methods Garbage collection & finalize () method.					
Unit 2: Java classes and Inheritance (6 Sessions)					
Class, instance, methods, overriding, final, super - keywords, Define class with instance variables and methods Object creation of class accessing member of class Argument passing, Constructors Method overloading, static data, static methods, static blocks this keyword, Nested & Inner classes Wrapper Classes String (String Arrays, String Methods, String Buffer), Super class & subclass, Abstract method					

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and classes Method overriding, final keyword super keyword Down casting and up casting Dynamic method dispatch.

Unit 3: Packages and Interfaces (4 Sessions)

Importing classes User defined packages Modifiers & Access control (Default, public, private, protected, private protected), Implementing interfaces, User defined interfaces Adapter classes.

Unit 4: Exception handling (4 Sessions)

Types of Exceptions try, catch, finally, throw, throws keywords, creating your own exception Nested try blocks Multiple catch statements User defined exceptions.

Unit 5: Java Input Output (6 Sessions)

Java IO package File Class Byte/Character Stream Buffered, reader / writer File reader / writer Print writer File Sequential / Random Serialization and de serialization.

Unit 6: Multithreading Programming (6 Sessions)

Java Thread Model, the main thread, creating a thread, creating multiple Threads, Thread priorities, Synchronization, Interthread Communication, Suspending, Resuming and Stopping thread.

Unit 7: GUI Programming (10 Sessions)

Components and Graphics Containers, Frames and Panels Layout, Managers, Border Layout, Flow Layout, Grid Layout, Card Layout AWT all Components Event Delegation Model, Event Source and Handlers, Event Categories, Listeners, adapters Anonymous Classes. Swing Features of swing, Model view Controller, Design pattern Swing components JButton, JRadio Button, JTextArea, JComboBox, JTable, JProgressBar, JSlider, JDialog

Unit 8: Java Collection Framework Collections (5 Sessions)

Overview The Collection Interfaces, Collection Interface, List Interface, Set Interface, Sorted Set Interface, The Collection Classes, Array List Class, Linked List Class, Hash Set Class.

Laboratory Exercises / Practical:

Every student will carry out minimum *Six Practical*/exercises based on the above units and submit the journal, which will be evaluated as part of continuous assessment.

Learning Resources:

Reference Books:

1. Horstman Cay, Cornell Gary, Core JavaTM2, Vol.1&2, Seventh Edition, Pearson education, 2004 Edition.
2. Herbert Schildt, The Complete Reference, Seventh Edition, Tata McGraw-Hill, 2007

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3. Java 2 Programming Black Book: AWT, Swing, Generics, XML, Sound, Animation, JDBC, Servlets, RMI, Threading, Sockets, Network and Java Beans by Steven Holzner, reprint, Dreamtech Press, 2008
4. Ivor Horton's Beginning Java 2, JDK 5 Ed, Wiley India by Ivor Horton, Jan. 2005

Supplementary Reading:

Web Resources:

1. <https://www.tutorialspoint.com> ,
2. www.javatpoint.com www.javatpoint.com

MOOCs: Online courses for self-learning

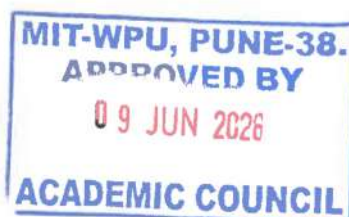
Courses by NPTEL and MIT Open Courseware etc.

https://onlinecourses.nptel.ac.in/noc22_cs47/preview

Pedagogy:

- Power point presentations
- Videos
- Demonstrations
- Interactive discussion.
- Practical based learning,
- Project based learning,

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COURSE STRUCTURE

Course Code	MCA40160				
Course Title	Advanced Database Management System				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3				3
Credits	3				3
Assessment Schema Code	TT1				
<u>Prerequisites:</u>					
Database Management System and Relational database management system, SQL					
<u>Course Objectives:</u>					
1. Understanding parallel database principles, distributed, XML, Objective oriented and mobile databases.					
<u>Course Outcomes:</u>					
After completion of this course students will be able to: 1. Design parallel and distributed databases. 2. To work on mobile, XML databases.					
<u>Course Contents:</u>					
Unit 1: Database System Architectures (6 Sessions)					
Database System Architectures Centralized and Client-Server Architectures, Server System Architectures, Parallel Systems, Distributed Systems.					
Unit 2: Parallel and Distributed Databases (12 Sessions)					
Introduction, I/O Parallelism, Inter and Intra Query Parallelism, Inter and Intra operation Parallelism, Distributed Data Storage, Distributed Transactions, Commit Protocols, Concurrency Control, Distributed Query Processing – Three Tier Client Server Architecture.					
Unit 3: Objected Oriented databases (9 Sessions)					
Concepts for Object Databases, Object Identity, Object structure and Type Constructors, Encapsulation of Operations, Methods and Persistence, Type and Class Hierarchies and Inheritance, Complex					

Objects, Object Database Standards, Languages and Design: ODMG Model – ODL – OQL – Object Relational and Extended – Relational Systems: Object Relational features in SQL/Oracle

Unit 4: XML databases (9 Sessions)

XML Data Model, XML Documents, DTD and XML Schema, XML Querying, Web Databases – JDBC – Information Retrieval – Data Warehousing – Data Mining

Unit 5: Mobile databases & Multimedia databases (9 Sessions)

Location and Handoff Management - Effect of Mobility on Data Management - Location Dependent Data Distribution - Mobile Transaction Models - Concurrency Control - Transaction Commit Protocols - Mobile Database Recovery Schemes Multidimensional Data Structures – Image Databases – Text/Document Databases- Video Databases – Audio Databases – Multimedia Database Design.

Laboratory Exercises / Practical:

Every student will carry out minimum *Six Practical*/exercises based on the above units and submit the journal, which will be evaluated as part of continuous assessment.

Learning Resources:

Reference Books:

1. AviSilberschatz, Henry F. Korth, S. Sudarshan, "Database System Concept", 4th Edition, 2001, ISBN: 0072283637, McGraw-Hill publications.
2. RamezElmasri, ShamkantNavathe, "Fundamental of Database Systems", 5th Edition, 2009, ISBN: 9780321369574, Pearson Education.
3. Alexis Leon, Mathews Leon, "Database Systems concept", 2002, ISBN: 9788125911654, Leon Vikas.

Supplementary Reading:

Web Resources:

1. <https://www.tutorialspoint.com>,
2. www.w3schools.com

MOOCs: Online courses for self-learning

Courses by NPTEL and MIT Open Courseware etc.

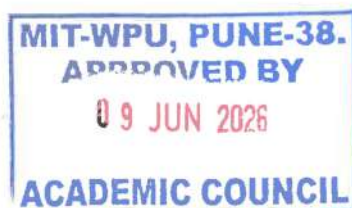
https://onlinecourses.nptel.ac.in/noc22_cs47/preview

Pedagogy:

- Power point presentations
- Videos
- Demonstrations
- Interactive discussion.
- Practical based learning,
- Project based learning,

COURSE STRUCTURE

Course Code	MCA40170				
Course Title	Python Programming				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	4				4
Credits	4				4
Assessment Schema Code	TT1				
<u>Prerequisites:</u>					
Basic programming skill, Algorithm					
<u>Course Objectives:</u>					
1. Fundamental understanding of the Python programming language. 2. To learn to use the Python programming language to solve problems of interest to them. It introduces the core programming basics including data types, operators, input/output, control structures, iterative and recursive constructs, compound data types, regular expression and program design with functions, python modules. 3. The course also discusses the object-oriented design using python.					
<u>Course Outcomes:</u>					
After completion of this course students will be able to: 1. To read, write and debug python programs to solve computational problems. 2. To select and use a suitable programming construct and data objects like lists, sets, tuples and dictionaries for solving a given problem. 3. To be proficient in the handling of functions and modules in python 4. To understand Object Oriented Programming using python					
<u>Course Contents:</u>					
Unit 1: Fundamentals of Python Programming (15 Lectures)					
Introduction to Python: Python installation, concept of data types; variables, Identifiers and keywords, Literals, Strings, Operators (Arithmetic operator, Relational operator, Logical operator, Boolean operator, Assignment operators, Membership operators (in and not in), Identity operators, Bit-wise operator, Increment or Decrement operator), comments in the program.					

Creation of a Python Program: Input and Output Statements, Control statements –Branching (if-else, if-elif-else), indentation in python, iteration (using for, while), Conditional Statement, exit function, Difference between break, continue and pass

Unit 2: Functions and Data Structure (15 Lectures)

Functions: Built in function (math, statistics), User defined functions: Defining Functions, arguments: positional, default, keyword, variable length arguments, scope of variables, return statement, recursion, importing (using import).

List – Adding Items to a List, Finding and Updating an Item, Nested Lists, Cloning Lists, Looping Through a List, sorting a List, List Concatenation, List Slices, List Methods, Mutability, Aliasing
Tuples: Creation, Accessing, Updating, Deleting Elements in a Tuple, Tuple Assignment, Tuple as Return Value, Nested Tuples, Basic Tuple Operations, Sets, Dictionary operations, Built-in Dictionary Functions & Methods

Unit 3: Regular Expressions and Modules (15 Lectures)

Regular Expressions – Concept of regular expression, various types of regular expressions, using match function.

Modules in python: use of keyword from, namespace, module aliasing, Introduction to python packages, pandas, numpy, scikitlearn, Matplotlib

Unit 4: Object Oriented Programming (15 Lectures)

Classes and Objects: Overview of OOP (Object Oriented Programming), Class Definition, Creating Objects, Instances as Arguments, Instances as return values, Built-in Class Attributes, Inheritance, Method Overriding, Data Encapsulation, Data Hiding, exception handling

Learning Resources:

Reference Books:

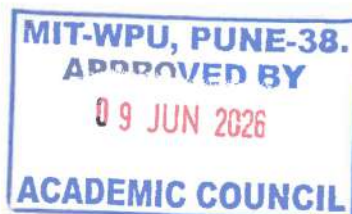
1. Downey, A.B., (2015), Think Python How to think like a Computer Scientist, 3rd edition.
2. Taneja, S. & Kumar, N., (2017), Python Programming- A Modular Approach. Pearson Education.
3. Brown, M. C. (2001). The Complete Reference: Python, McGraw Hill Education.
4. Dromey, R. G. (2006), How to Solve it by Computer, Pearson Education.
5. Guttag, J.V.(2016), Introduction to computation and programming using Python. MIT Press.
6. Liang, Y.D. (2013), Introduction to programming using Python. Pearson Education.

Supplementary Reading:

Web Resources:

Weblinks:

1. <https://www.javatpoint.com>



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2. <https://www.tutorialspoint.com/asp.net/index.htm>
3. www.W3schools.com
4. <https://www.programiz.com/python-programming>

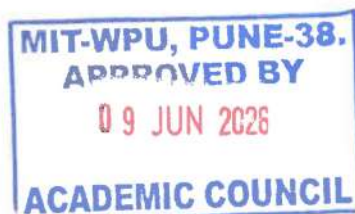
MOOCS: Spoken Tutorial, Swayam Courses

Pedagogy:

1. Practical learning
2. Problem solving assignment.
3. Conceptual and Contextual learning
4. PPT Presentation & Interactive discussion

Sample Assignments on Python Programming for Tutorial	
1	➤ Demonstrate about Fundamental Data types in Python Programming ➤ Write a program to perform different arithmetic operations on numbers in python. ➤ Write a program to create, concatenate and print a string and accessing sub-string from given string. ➤ Write a Python program to Remove all duplicates from a given string in Python. ➤ Using a for loop, write a program that prints the decimal equivalents of 1/2, 1/3, 1/4, 1/10 ➤ Write a program using a while loop that asks the user for a number and prints a countdown from that number to zero. ➤ Create a menu driven Python program using user defined functions to implement a calculator to perform: (a) Basic arithmetic operations (b) $\log_{10}(x)$, $\sin(x)$, $\cos(x)$
2	➤ Write a program which makes use of the function to display all such numbers which are divisible by 7 but are not a multiple of 5, between 1000 and 2000. ➤ Write a python program which contains user defined functions as a 'module' to calculate area, perimeter or surface area, volume for various shapes like square, cube, circle, cylinder. The user defined functions should accept the values for calculation as parameters and calculated values should be returned. Import the module and use appropriate functions. ➤ Demonstrate the following functions/methods which operates on lists in Python with suitable

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	examples: len () ,count () ,index () ,append () ,insert () ,extend () ,remove () , pop () ,reverse () ,copy () ➤ Write a python program to read a list of 'n' integers (positive and negative) and create two new lists one having all positive numbers and the other having all negative numbers from the given list. Print all three lists. ➤ Demonstrate the following functions/methods which operates on tuples in Python with suitable examples: len () ,count () ,index () ,sorted () ,min () , max () , cmp () , reversed () ➤ Demonstrate the following functions/methods which operates on dictionary in Python with suitable examples: clear () , get () , pop () , copy () , pop () ,keys () , values () ,items () ➤ Create a Python program to create a dictionary which has a record of a student information: Admission number, Roll Number, Name and Marks. Display information on the basis of Admission number. ➤ Write a Python script to sort (ascending and descending) a dictionary by value.
3	➤ Create a numpy array to demonstrate basic array characteristics and Apply basic operations +,- ,*/ and find the transpose of matrix. ➤ Write a Python program to visualize a dataset by displaying multiple types of charts using Matplotlib. ➤ Write a Python program to load the iris data from a given csv file into a dataframe and print the shape of the data, type of the data and first 3 rows. ➤ Write a python program to count NaN or missing values in Pandas DataFrame. ➤ Write a NumPy program to convert Centigrade degrees into Fahrenheit degrees. Centigrade values are stored in a NumPy array ➤ Demonstrate the following in-built functions to use Regular Expressions very easily in our applications. finditer () , match () , fullmatch () ,search () ,findall () , sub () , split ()
4	➤ Write a Python program to create a person class. Include attributes like name, country and date of birth. Implement a method to determine the person's age. ➤ Write a python program to implement Multiple inheritance and Method overriding. ➤ Write a python program to implement try, catch, finally

COURSE STRUCTURE

Course Code	MCA40180				
Course Title	Lab on Java Programming				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours			2		2
Credits			1		1
Assessment Schema Code	PR				
Prerequisites:					
Basic programming knowledge of C and C++, concept of object-oriented programming.					
Course Objectives:					
4. To enable the students to understand the core principles of the Java Language and use visual tools to produce well designed, effective applications and applets.					
5. To make students understand and learn the basic Java programming.					
6. To make students build desktop-based applications using Java					
Course Outcomes:					
After completion of this course students will be able to:					
4. Create Java application using object-oriented concepts.					
5. Understand and apply the concepts of exception handling.					
6. Apply the concepts of multithreading, java collections and file handling.					
Course Contents:					
Sample Lab Assignments					
1. Write a program to check whether a given number is prime or not.					
2. Write a program to accept range from command line argument & print the perfect numbers from the given range.					
3. Write a program to calculate and print the first n Fibonacci numbers. (Accept n from command line).					
4. Write a program to check whether a given text or number is palindrome or not.					
5. Write a program to print the following outputs using for loops.					
1					
12					
123					

*ep
B.102*



- 1234
12345
6. Write a program to accept employee information such as Emp_id, Name, Designation and Salary using parameterized constructor and display the same.
 7. Create a class Account with members account no., acc type, cust name, balance & methods deposit, withdraw & show balance. Write a main class to accept all the details & display all details.
 8. Write a program to accept five elements in an array and find the max and min from given numbers.
 9. Write a Java program can contain two classes: Computer and Laptop. Both classes have their own constructors and a method. In main method create object of two classes and call their methods.
 10. Create person class with data members as person_id & name. Derive two classes Student & faculty from it. The fields of Student are course name & fees paid. The fields of faculty are subject name & number of years' experience. Use proper method to accept values & override display method. (Using parameterized constructor)
 11. Create a class mobile containing company name, mobile number & cost and write necessary member functions for the following:
 12. Search the mobile number with given name.
 13. Search the name with given telephone number. (Use method overloading)
 14. Write a Java program to create a package for Book details giving Book Name, Price and Author name and import the created package using importing package where we can create the object of classes in this package.
 15. Write a program to create interface A in this interface we have two methods display() which will display variables of class and print_pattern() which will print pattern. Implements this interface in another class named MyClass.
 16. Write a java program to create two packages. In first package create employee class with employee number, name & salary. In second package create a class manager which is subclass of employee class having fields as department name & incentive. Create main class which will import these packages & display gross salary.
 17. Accept two numbers a & b from command line argument & print output as a/b and handle all possible system defined exceptions.
 18. Write a class Driver with attributes vehicle no, name & age. Initialize values through parameterized constructor. If age of driver is less than 18 then generate user-defined exception "Age Not Within the Range".
 19. Write a program to demonstrate use of user-defined exception, the CheckingAccount class contains a withdraw () method that throws an Insufficient Funds Exception.
 20. Write a program to implement the concept of threading by extending Thread Class and also by implementing interfaces.
 21. Write a program to set the priority of two above threads and check which thread executes first.
 22. Write a program to print name, priority of a thread and change name of current to java thread and display the details of current thread.

Laboratory Exercises / Practical:

Every student will carry out minimum *Six Practical*/exercises based on the above units and submit the journal, which will be evaluated as part of continuous assessment.

Learning Resources:

Reference Books:

1. Horstman Cay, Cornell Gary, Core JavaTM2, Vol.1&2, Seventh Edition, Pearson education, 2004 Edition.
2. Herbert Schildt, The Complete Reference, Seventh Edition, Tata McGraw-Hill, 2007
3. Java 2 Programming Black Book: AWT, Swing, Generics, XML, Sound, Animation, JDBC, Servlets, RMI, Threading, Sockets, Network and Java Beans by Steven Holzner, reprint, Dreamtech Press, 2008
4. Ivor Horton's Beginning Java 2, JDK 5 Ed, Wiley India by Ivor Horton, Jan. 2005

Supplementary Reading:

Web Resources:

1. <https://www.tutorialspoint.com> ,
2. www.javatpoint.com www.javatpoint.com

MOOCs: Online courses for self-learning

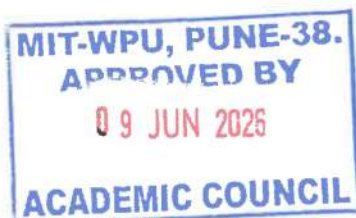
Courses by NPTEL and MIT Open Courseware etc.

https://onlinecourses.nptel.ac.in/noc22_cs47/preview

Pedagogy:

- Power point presentations
- Videos
- Demonstrations
- Interactive discussion.
- Practical based learning,
- Project based learning,

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COURSE STRUCTURE

Course Code	MCA40190				
Course Title	Lab on Advanced Database Management System				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours			2		2
Credits			1		1
Assessment Schema Code	PR				
<u>Prerequisites:</u>					
Database Management System and Relational database management system, SQL					
<u>Course Objectives:</u>					
1. Understanding parallel database principles, distributed, XML, Objective oriented and mobile databases.					
<u>Course Outcomes:</u>					
After completion of this course students will be able to:					
1. Design parallel and distributed databases.					
2. To work on mobile, XML databases.					
<u>Course Contents:</u>					
Sample Lab Assignments					
1. Write down Stored Procedure for inserting, updating values in the table					
2. Write down stored procedures to accept input values as a parameter and update values of the tables					
3. Write down DML trigger to raise the error while inserting duplicate value in the table					
4. Write down DML trigger to raise the error violating Check Constraints					
5. Write down DML trigger to raise the error when user deletes more than records from table					
6. Create index and measure the performance of query on the table					
7. Compare results before and after indexing by considering any sample table					

8. Write down the stored procedure for selecting the first five records and copy those five records in another table
9. Write down a function which prints higher salaried person from table by inputting his empid and name
10. Write down the function which selects part name (substring, like surname, firstname or middle name and copy that part in another table's column)

Laboratory Exercises / Practical:

Every student will carry out minimum *Six Practical*/exercises and submit the journal, which will be evaluated as part of continuous assessment.

Learning Resources:

Reference Books:

1. AviSilberschatz, Henry F. Korth, S. Sudarshan, "Database System Concept", 4th Edition, 2001, ISBN: 0072283637, McGraw-Hill publications.
2. RamezElmasri, ShamkantNavathe, "Fundamental of Database Systems", 5th Edition, 2009, ISBN: 9780321369574, Pearson Education.
3. Alexis Leon, Mathews Leon, "Database Systems concept", 2002, ISBN: 9788125911654, Leon Vikas.

Supplementary Reading:

Web Resources:

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Pedagogy:

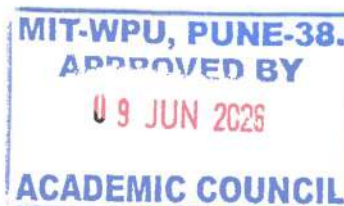
- Power point presentations
- Videos
- Demonstrations
- Interactive discussion.
- Practical based learning,
- Project based learning,

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COURSE STRUCTURE

Course Code	DCA40010				
Course Title	Research Methodology				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	4				4
Credits	4				4
Assessment Schema Code	TT1				
<u>Prerequisites:</u>					
Basic Understanding of Research Concepts and statistical tools, Flowchart Preparation, Algorithm development, and core techniques.					
<u>Course Objectives:</u>					
Students will be able to:					
1. Demonstrate through the literature, Methodologies and methods that will work well for their own research.					
2. Create, understand, and share new information through publishable original research that passes peer review.					
3. Present and defend innovative research findings that advance a field or pertinent area of professional practice.					
4. Evaluate current challenges, research, and cutting-edge scholarship in the field critically and creatively.					
5. To prepare an ethical scientific research report.					
<u>Course Outcomes:</u>					
After completion of the Course student should be able to:					
1. Understand the critical importance of methodology in research.					
2. Analyse and comprehend the collected data using appropriate statistical tools and techniques.					
3. Identify and select appropriate techniques and strategies for developing the research framework.					
4. Understand and write research papers that are methodologically sound, and conceptually cohesive.					

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Course Contents:

Unit 1: Introduction (12 Sessions)

Meaning and objective of Research Methodology, Motivation in research, Types of research, Research Approaches, Significance of research, Research Methods versus Methodology, Research Process, Criteria of Good Research, Nature of Computer Science (CS) research, Defining the Research Problem: Research Problem, Selecting the Problem, Necessity of Defining the Problem, Technique Involved in Defining a Problem,

Unit 2: Review of Literature (12 Sessions)

Basic Concept and its Need, Significance of literature review, Method /Approach for literature review and Documentation, Gap Analysis, Source for Literature: books -journals – proceedings - thesis and dissertations - unpublished items. Online Searching: Database – SciFinder – Scopus - Science Direct - Searching research articles - Citation Index - Impact Factor - H-index, etc

Unit 3: Research Design: (12 Sessions)

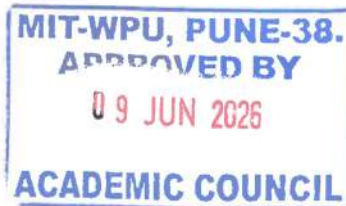
Meaning of Research Design, Need for Research Design, Features of a Good Design, Important Concepts Relating to Research Design, Different Type Research Designs, Basic Principles of Experimental Designs and Conclusion, Types of Data collection, Methods of Data Collection: Collection of Primary Data, Observation Method, Interview Method, Collection of Data through Questionnaires, Collection of Data through Schedules, Difference between Questionnaires and Schedules.

Unit 4: Modelling Analysis and Discussion: (12 Sessions)

Structured and Unstructured Data, Scales of Measurement, Population, and Sample, Descriptive Statistics, Data Visualization. Probability and random variables, hypothesis testing, Data Analytics: Elements of Association, Clustering, and Classification. Model and Model Building: Classification of Models, Variable Relationships, Model Boundary, and Predictive and Prescriptive Models. Types of analysis, Classification and Tabulation, Interpretation, Hypothesis Testing, Findings and limitation of Research, Future Enhancement

Unit 5: Research Ethics and Publication (12 Sessions)

Research Ethics & Its Importance, Codes & Policies, Conducting & Publishing Research Ethically,



Research Misconduct, Proposal for ethical clearance, footnotes & bibliographies, intellectual property: copyright & plagiarism, publishing/editing/ethical issues/peer review, submission of research papers, writing skills structured paper (Report writing and evaluations)

Learning Resources:

Reference Books:

1. Kitsakorn Locharoenrat, Research Methodologies for Beginners, Pan Stanford Publishing Pte. Ltd., Singapore, 2017.
2. Business Research Methods- Donald Cooper & Pamela Schindler, TMGH, 9th editions.
3. C.R. Kothari and Gaurav Garg—Research Methodology, Fourth Edition, New Age International Publishers. 2020.
4. 'The Statistical Analysis of Experimental Data' by, John Mandel, ISBN: 0486646661, ISBN: 9780486646664
5. Donald R Cooper and Pamela S Schindler, Business Research Methods, Tata McGraw Hill Publishing Company Ltd.
6. Sinha, S.C. and Dhiman, A.K., 2002. Research Methodology, Ess Ess Publications. 2 volumes.

Web Resources:

1. <https://www.udemy.com/course/research-methods-and-statistics-advanced/>
2. <https://www.coursera.org/learn/research-methods>
3. https://onlinecourses.nptel.ac.in/noc22_gc08/preview
4. <https://www.udemy.com/topic/research-methods/>

Pedagogy:

- Problem-Based Learning (PBL),
- Case Studies,
- Group Discussions,
- Research Proposal Development,
- Statistical Software Workshops,
- Research Journals

SEMESTER-II

MIT-WPU, PUNE-38.
APPROVED BY
09 JUN 2026
ACADEMIC COUNCIL

COURSE STRUCTURE

Course Code	MCA40200				
Course Title	Advanced Java				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3				3
Credits	3				3
Assessment Schema Code	TT1				
<u>Prerequisites:</u>					
Object Oriented Concepts, Basic Java Programming, Database and Web Technology concepts					
<u>Course Objectives:</u>					
1. Explore advanced topic of Java programming for solving problems					
2. Be able to put into use the advanced features of the Java language to build and compile robust enterprise- grade applications					
3. Provide a sound foundation to the students on the concepts, precepts and practices, in a field that is of immense concern to the industry and business					
4. Enhance knowledge to manipulate and store data					
<u>Course Outcomes:</u>					
After completion of this course students will be able to:					
1. To learn and implement the concepts of advanced java like Servlet, JSP, Spring and Hibernate framework.					
2. To understand and apply the concepts of RMI and Java Beans.					
3. To build the web application using the JDBC and other related concepts.					
<u>Course Contents:</u>					
Unit 1: Database Programming Using JDBC (7 Sessions)					
Java database connectivity, JDBC Architecture, JDBC API, Types of JDBC drivers, Steps to create JDBC Application, writing first JDBC applications, Types of statement objects (Statement, Prepared					

Statement & Callable Statement), Types of result set, Result Set Metadata, Inserting and updating records, JDBC and AWT, Connection pooling.

Unit 2: Networking with Java (7 Sessions)

Networking basics – Sockets, port – Proxy servers, java.net – networking classes and interfaces, Implementing TCP/IP based Server and Client, Datagrams – Datagram packet, Datagram server and client, URL connections, Multithreaded Chat Server, Multithreaded socket Programming.

Unit 3: Servlets (8 Sessions)

Introduction, Servlet vs CGI, Servlet API Overview, Servlet Life Cycle, Coding: Writing & running simple servlet, Generic servlet, HttpServlet, ServletConfig, ServletContext, writing servlet to handle Get & Post methods, reading use request data, Session tracking in servlets, Servlets & JDBC, writing thread safe servlet Note: Apache Tomcat server is used at server side.

Unit 4: JSP (8 Sessions)

Why JSP? JSP Directives, writing simple JSP page, Scripting Elements, Default Objects in JSP, JSP Actions Managing Sessions using JSP, JSP with beans, JSP & Databases, Error Handling in JSP, Introduction to custom tag, JSP with JDBC Note: Apache Tomcat server is used at server side.

Unit 5: Spring Framework (9 Sessions)

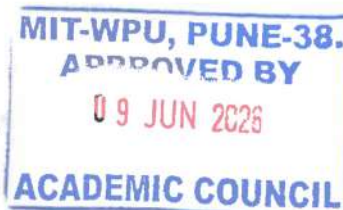
Introduction, Architecture, Spring bean wiring, Spring IOP and Dependency Injection, AOP with spring, Spring with database.

Unit 6: Hibernate Framework (6 Sessions)

Introduction, Object Relation Mapping (ORM), Hibernate Architecture, Environment Setup, Configuration.

Laboratory Exercises / Practical:

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Every student will carry out minimum *Six Practical/exercises* based on the above units and submit the journal, which will be evaluated as part of continuous assessment.

Learning Resources:

Reference Books:

1. Java Complete Reference Patric Naughton, Herbert Schildt, TMH, 11th Edition, 2018
2. Complete Reference- J2EE by Jim Keogh, 1st Edition, TMH, 2017
3. Inside Servlets by Dustine R. Callway, Pearson Education, 1st edition, 2009
4. JDBC, Servlet and JSP, Black Book, Santosh Kumar K. Dremtech publication 2nd Ed. Jan 2016
5. Professional JSP Wrox press 2nd Edition 1-Jan 2001.
6. Spring Persistence with Hibernate, Ahmad Seddighi, Packt Publishing Limited, 1st edition, 2009.
7. Java unleashed by Michael Morrison, Sams.net publication, 1st Edition, 1997
8. Spring and hibernate, Santosh Kumar K., 1st edition, McGraw Hill Education, 2009

Supplementary Reading:

Web Resources:

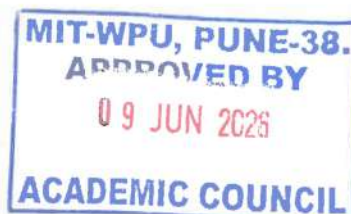
Weblinks:

1. <https://www.javatpoint.com>
2. <https://www.tutorialspoint.com/asp.net/index.htm>
3. www.W3schools.com
4. <https://www.udemy.com/course/advanced-java-programming/>
5. <https://www.edureka.co/blog/advanced-java-tutorial>

MOOCS: Spoken Tutorial, Khan Academy, Swayam Courses on Advanced Java

Pedagogy:

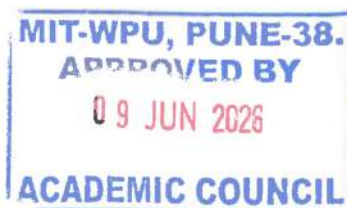
1. Practical learning
2. Problem solving assignment.
3. Conceptual and Contextual learning
4. PPT Presentation & Interactive discussion



COURSE STRUCTURE

Course Code	MCA40210				
Course Title	Data Structures and Algorithms				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3				3
Credits	3				3
Assessment Schema Code	TT1				
<u>Prerequisites:</u>					
Procedural Languages C, Object Oriented Paradigm using C++, Basic Knowledge about Algorithms					
<u>Course Objectives:</u>					
1. To make the students understand about basic concepts and importance of Data structures.					
2. To make the students familiar with different algorithms like Stack, Queues, Linked Lists, Trees, Graphs etc.					
<u>Course Outcomes:</u>					
After completion of this course students will be able to:					
1. Understand and apply data structures.					
2. Use linear and non-linear data structures like stacks, queues, linked list etc.					
3. Learn understanding of tree, graph data structures.					
<u>Course Contents:</u>					
Unit 1: Introduction to Data Structures and Algorithms: (7 Sessions)					
Algorithmic Notation: Format Conventions, Statement and Control Structures, Time and Space Analysis: Data types and Abstract data types, Types of Data structures; Primitive, Non primitive, Linear and Nonlinear Data structures.					
Unit 2: Array: (7 Sessions)					

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Storage representation, operations and applications (Polynomial addition and subtraction) Stack: operations and applications (infix, postfix and prefix expression handling), Queue: operations and applications, Circular Queues: operations and applications, Concept of Double ended Queue and Priority Queue, Linked representation of stack and queue.

Unit 3: Searching: (7 Sessions)

Linear Search and Binary Search (array/binary tree). **Sorting:** General Background, Sorting Techniques: Bubble Sort, Insertion Sort, Selection Sort, Quicksort, Merge sort, Heapsort and Radix Sort.

Unit 4: Linked Lists: (6 Sessions)

Operations and Applications of Linear linked list (Polynomial addition and subtraction), Circular linked list and doubly linked list.

Unit 5: Trees: (6 Sessions)

Binary Trees, Binary Tree: Representations, Operations (insert/delete), Traversal (in order, preorder, post order, level order), Threaded Binary Tree, Search Trees: AVL Tree, single and double rotations, B-Trees: insertion and deletion, Introduction to B+ and B* Trees.

Unit 6: Graphs and Their Applications: (6 Sessions)

Representation (Matrix/Adjacency) and Traversal (Depth First Search/Breadth First Search), Spanning Trees, Minimal Spanning Tree (Prim's and Kruskal's algorithm), Shortest Paths and All Pair Shortest Path, Dijkstra's, Floyd-Warshall Algorithms.

Unit 7: Hash Table: (6 Sessions) (6 Sessions)

Hash Function, Collision and its Resolution, Separate Chaining, Open Addressing (linear probing, quadratic probing, double hashing), Rehashing, Extendible Hashing.

Laboratory Exercises / Practical:

Every student will carry out minimum *Six Practical*/exercises based on the above units and submit the journal, which will be evaluated as part of continuous assessment.

Learning Resources:

Reference Books:

1. An Introduction to Data Structures with Application by Tremblay, J. & Sorenson, P.G., Tata McGraw Hill India, 2nd edition, 2001,
2. Data Structures using C and C++ by Langsam, Y., Augenstein, M. J. & Tenenbaum A.M, 2nd Edition, Pearson Education, 2015
3. Data Structures using C by Balagurusamy, E., 1st Edition, McGraw Hill Education, 2013
4. Data Structures and Algorithm Analysis in C by Weiss, M.A., 2nd Edition, Pearson India, 2002
5. Data Structures and Algorithm Analysis in C by Horowitz, E.; Sartaj S. & Mehta, D., X Universities Press, 1st edition, 2008
6. Data Structures & Algorithms in Java by Lafore, R., 2nd Edition, Pearson India, 2008
7. Data Structures and Program Design in C by Kruse, R., Tondo, C.L., Leung B., & Mogalla S, Pearson India, 2nd edition, 2006.

Supplementary Reading:

Web Resources:

MOOCS: Spoken Tutorial, Khan Academy, Swayam Courses

Weblinks:

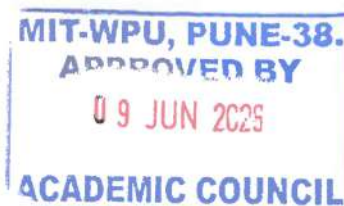
1. <https://www.geeksforgeeks.org/best-data-structures-and-algorithms-courses/>
2. <https://plainenglish.io/blog/5-platforms-to-learn-algorithms-and-data-structures-3152238de56c#1-hackerrank>

Pedagogy:

1. Participative learning
2. Discussions
3. Problem solving
4. Assignments

COURSE STRUCTURE

Course Code	MCA40220				
Course Title	Statistics and Probability				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	4				4
Credits	4				4
Assessment Schema Code	TT1				
<u>Prerequisites:</u>					
Basic Maths and Statistics Knowledge					
<u>Course Objectives:</u>					
1. Understand the mathematical underpinnings of probability. 2. To learn the basics of business decision-analysis 3. To understand the process associated with statistical decisions, defining, and formulating problems, analysing the data, and using the results in decision making.					
<u>Course Outcomes:</u>					
After completion of this course students will be able to:					
1. Demonstrate concepts of business statistics (such as measures of central tendency, dispersion, correlation, regression analysis and time series analysis) 2. based on the acquired knowledge to interpret the meaning of the calculated statistical indicators 3. Demonstrate concept of Random Variable for solving practical problems in business world					
<u>Course Contents:</u>					
Unit 1: Measures of Central Tendency and Dispersion (12 Sessions)					
Introduction, Objectives of statistical average, Requisites of a Good Average, Statistical Averages - Arithmetic mean - Properties of arithmetic mean - Merits and demerits of arithmetic mean. Median - Merits and demerits of median. Mode - Merits and demerits of mode, Geometric Mean, Harmonic Mean, Appropriate Situations for the Use of Various Averages, Positional Averages. Dispersion – Range – Quartile deviations, Mean deviation, Standard Deviation -Properties of standard deviation Coefficient of Variance					

Unit 2: Simple Correlation and Regression (10 Sessions)

Introduction, Correlation - Causation and Correlation - Types of Correlation - Measures of Correlation - Scatter diagram – Karl Pearson's correlation coefficient - Properties of Karl Pearson's correlation coefficient - Factors influencing the size of correlation coefficient Multiple Correlations, Regression - Regression analysis - Regression lines - Regression coefficient. Standard Error of Estimate, Multiple Regression Analysis.

Unit 3: Counting Principles (8 Sessions)

Addition and Multiplication Principles, Permutations of n Objects, Circular Permutation, Permutation with repetitions, Combinations and combinatorial identities, Binomial and Multinomial Theorems and its applications, Derangements – Nothing in its right place.

Unit 4: Introduction To Probability (12 Sessions)

Trials, Events, Sample Space – Types and Examples, Mathematical Probability, Axioms of Probability, Some elementary theorems in probability, Independent and Dependent Events, Conditional Probability, Baye's Theorem.

Unit 5: Random Variables and Mathematical Expectation (12 Sessions)

Random Variable – Discrete and Continuous, Probability Distribution of a Random Variable, Probability Mass Function, Probability Density Function, Distribution Functions, Mathematical Expectation of Probability Distribution, Theorems, Calculation of Mean and Variance using Mathematical Expectation, Moment Generating Functions and Cumulant Generation Functions.

Unit 6: Forecasting and Time Series Analysis (6 Sessions)

Introduction to Types of forecasts, Timing of forecast, Quantitative and Qualitative Forecasting Methods. Introduction to Time Series Analysis Components of Time Series

Reference Books:

1. Probability & Random Process by T. Veerarajan
2. Fundamentals of Mathematical Statistics by S. C. Gupta and V. K. Kapoor
3. Statistical Methods by S. P. Gupta

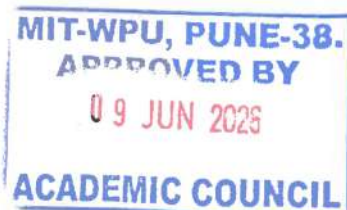
Supplementary Reading:

Web Resources:

1. [khanacademy.com](https://www.khanacademy.com)

Pedagogy:

Participative learning, discussions, problem solving, assignments, Tutorials, Lab assignments



COURSE STRUCTURE

Course Code	MCA50170				
Course Title	Data Science and Analytics				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	4				4
Credits	4				4
Assessment Schema Code	TT1				
<u>Prerequisites:</u>					
Knowledge of RDBMS, MySQL, Python					
<u>Course Objectives:</u>					
1. Explain the foundational principles and the importance of data science, as well as its relationship with information science. 2. Apply descriptive and inferential statistical methods to summarize, analyze, and interpret data for decision-making. 3. Solve optimization problems using gradient-based and non-gradient techniques relevant to data science scenarios. 4. Perform data acquisition, preprocessing, and visualization, using appropriate techniques and tools to represent and validate datasets effectively. 5. Utilize open-source software and programming libraries to implement data science solutions.					
<u>Course Outcomes:</u>					
1. To introduce the fundamental concepts and need for data science and its application in solving real-world problems. 2. To develop an understanding of statistical methods and their role in data analysis, including hypothesis testing and statistical modelling.					

3. To explore optimization techniques for data science applications, focusing on both constrained and unconstrained scenarios.
4. To provide hands-on skills in data acquisition, preprocessing, and visualization, enabling effective data handling and representation.
5. To familiarize students with open-source tools and programming libraries essential for data science and machine learning applications.

Course Contents:

Unit I: Introduction to Data Science (8 Sessions)

Basics and need of Data Science, Applications of Data Science, Data explosion, Relationship between Data Science and Information Science, Introduction to Data Analytics Lifecycle, statistics and optimization from a data science perspective; Structured thinking for solving data science problems, Structured and unstructured data

Unit II: Statistical Inference (12 Sessions)

Descriptive statistics, Statistical Features, summarizing the data, outlier analysis, Understanding distributions and plots, Univariate statistical plots and usage, Bivariate and multivariate statistics, Dimensionality Reduction, Over and Under Sampling, Bayesian Statistics, Statistical Modelling for data analysis, Basics and need of hypothesis and hypothesis testing, Pearson Correlation, Sample Hypothesis testing, Chi-Square Tests, t-test.

Unit III: Optimization (15 Sessions)

Unconstrained optimization; Necessary and sufficiency conditions for optima; Gradient descent methods; Constrained optimization, KKT conditions; Introduction to non-gradient techniques; Introduction to least squares optimization

Unit IV: Data Handling and Visualization (15 Sessions)

Regression and Classification, Data Acquisition, Data Pre-processing and Preparation, Data Quality and Transformation, Handling Text Data; Introduction to data visualization, Visualization workflow: describing data visualization workflow, Visualization Periodic Table; Data Abstraction -Analysis: Four

Levels for Validation- Task Abstraction - Analysis: Four Levels for Validation Data Representation:

chart types: categorical, hierarchical, relational, temporal & spatial

Unit V: Data Science Tools and Techniques (10 Sessions)

Overview and Demonstration of Open-source tools such as R, Octave, Scilab. Python libraries: SciPy and sci-kitLearn, PyBrain, Pylearn2; Weka.

List of Lab Assignments:

1. Perform the following operations using Python on any open-source dataset (e.g., data.csv)

- Locate an open-source data from the web (e.g., <https://www.kaggle.com>). Provide a clear
- Import all the required Python Libraries.
- Description of the data and its source (i.e., URL of the web site).
- Load the Dataset into pandas dataframe.
- Data Preprocessing: check for missing values in the data using pandas isnull(), describe()
- function to get some initial statistics. Provide variable descriptions. Types of variables etc.
- Check the dimensions of the data frame. Data Formatting and Data Normalization: Summarize the types of variables by checking the data types (i.e., character, numeric, integer, factor, and logical) of the variables in the data set. If variables are not in the correct data type, apply proper type conversions. Turn categorical variables into quantitative variables in Python.

2. Data Wrangling II

Create an “Academic performance” dataset of students and perform the following operations using Python.

- Scan all variables for missing values and inconsistencies. If there are missing values and/or inconsistencies, use any of the suitable techniques to deal with them.
- Scan all numeric variables for outliers. If there are outliers, use any of the suitable techniques to deal with them.

3. Perform the following operations on any open-source dataset (e.g., data.csv)

- Provide summary statistics (mean, median, minimum, maximum, standard deviation) for a dataset (age, income etc.) with numeric variables grouped by one of the qualitative (categorical) variable. For example, if your categorical variable is age groups and quantitative variable is income, then provide summary statistics of income grouped by the age groups. Create a list that contains a numeric value for each response to the categorical variable.
- Write a Python program to display some basic statistical details like percentile, mean, standard deviation etc. of the species of 'Iris-setosa', 'Iris-versicolor' and 'Iris-versicolor' of iris.csv dataset.

4. Data Analytics I

- Create a Linear Regression Model using Python/R to predict home prices using Boston Housing Dataset (<https://www.kaggle.com/c/boston-housing>). The Boston Housing dataset contains information about various houses in Boston through different parameters. There are 506 samples and 14 feature variables in this dataset. The objective is to predict the value of prices of the house using the given features.

5. Data Analytics II

- Implement logistic regression using Python/R to perform classification on Social_Network_Ads.csv dataset.
- Compute Confusion matrix to find TP, FP, TN, FN, Accuracy, Error rate, Precision, Recall on the given dataset.

6. Data Analytics III

- Implement Simple Naïve Bayes classification algorithm using Python/R on iris.csv dataset.
- Compute Confusion matrix to find TP, FP, TN, FN, Accuracy, Error rate, Precision, Recall on the given dataset.

7. Data Visualization I

- Use the inbuilt dataset 'titanic'. The dataset contains 891 rows and contains information about the passengers who boarded the unfortunate Titanic ship. Use the Seaborn library to see if we can find any patterns in the data.
- Write a code to check how the price of the ticket (column name: 'fare') for each passenger is distributed by plotting a histogram.

8. Data Visualization II

- Use the inbuilt dataset 'titanic' as used in the above problem. Plot a box plot for distribution of age with respect to each gender along with the information about whether they survived or not. (Column names : 'sex' and 'age';)
- Write observations on the inference from the above statistics.

9. Data Visualization III

- Download the Iris flower dataset or any other dataset into a DataFrame. (e.g., <https://archive.ics.uci.edu/ml/datasets/Iris>). Scan the dataset and give the inference as:
 - List down the features and their types (e.g., numeric, nominal) available in the dataset.
 - Create a histogram for each feature in the dataset to illustrate the feature distributions.
 - Create a boxplot for each feature in the dataset.
 - Compare distributions and identify outliers.

Learning Resources:

Reference Books:

1. R. V. Hogg, J. W. McKean and A. Craig, Introduction to Mathematical Statistics, 8th Ed., Pearson Education India, 2019.
2. Avrim Blum, John Hopcroft, Ravindran Kannan, "Foundations of Data Science", Cambridge University Press, 2020.
3. Ani Adhikari and John DeNero, "Computational and Inferential Thinking: The Foundations of Data Science", GitBook, 2019.
4. Cathy O'Neil and Rachel Schutt, "Doing Data Science: Straight Talk from the Frontline", O'Reilly Media, 2013.
5. Hossein Pishro-Nik, "Introduction to Probability, Statistics, and Random Processes", Kappa Research, LLC, 2014.

Supplementary Reading:

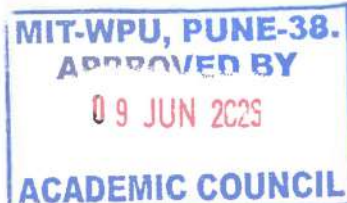
Web Resources:

MOOCS: NPTEL, Coursera

Pedagogy:

- Participative learning,
- discussions,

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- Program writing,
- experiential learning through practical problem-solving,
- assignment,
- PowerPoint presentation

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MIT-WPU, PUNE-38.
APPROVED BY
09 JUN 2025
ACADEMIC COUNCIL

COURSE STRUCTURE

Course Code	MCA50180				
Course Title	Mobile Application Development-iOS Programming				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	4				4
Credits	4				4
Assessment Schema Code	TT1				
<u>Prerequisites:</u>					
Object Oriented Concepts, Basic Programming Knowledge in Java or C++, Mobile Technologies.					
<u>Course Objectives:</u>					

Course Contents:

Unit I: Understanding Programming Standards (8 Sessions)

What are programming standards, Importance of programming standards, Benefits of using, programming standards, Programming Standards in detail, Good programming habits., Unit Testing and its importance.

Unit II: Introduction to Swift Language (15 Sessions)

Variables & Constants, Introduction to functions (methods), Arrays, Dictionaries, Data, Date and other basic data types, Enums, structures, closures, For, If, switch statements, Object oriented concepts with SWIFT, Type check, AnyObject, Any, Protocols, Extensions, Error handling, Working with classes

Unit III: Working with Xcode (5 Sessions)

Introduction to XCODE, COCOA touch framework, iOS application architecture, Application lifecycle

Unit IV: Introduction to view controllers and Views (18 Sessions)

View Controllers, view, view lifecycle, Basic Controls – Label, Buttons, Text field, image View, Table view with default cells and customized cells, Collection view with default cells and customized cells, Picker view, Date picker, scroll view, navigation and Tab bar controller, Understanding Interface builder, XIB files, Creating outlets and Actions, Handling touch and gesture events, Segment and Page control, switch view, UIAlertView

Unit V: Communicating with other view Controllers (8 Sessions)

Using storyboard to design the app, using segues, Introduction to methods used for passing data from one to another view controller.

Unit VI: Auto-Layout (6 Sessions)

Introduction to Auto-layout, Working with constraints, Using constraints to change view sizes, Programmatically control the views

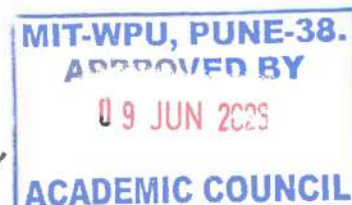
Laboratory Exercises / Practical:

Every student will carry out minimum *Six Practical*/exercises based on the above units and submit the journal, which will be evaluated as part of continuous assessment.

Sample Lab Assignments

Lab Assignments on Mobile application development

1	To print first Swift Code: Number and Cubes.
2	To reverse a function in Swift language.
3	Write a Program to print the value of a variable inside a text, place it in parentheses, and insert a backslash just prior to the opening parenthesis using swift language.
4	Write a Program to print constant and variables using swift language.
5	Write a Program to implement Swift Function Parameters and Return Values.
6	Write a Program to Swift break Statement by a program to find first 5 multiples of 6.
7	Write a Program to understand Nested Loops in Swift by a program to display 7 days of 2weeks.
8	Write a Program to understand Swift while and repeat while Loop by a program to display numbers from 1 to 5.
9	Write a program for repeat while in swift language.
10	Write a program for switch-case in swift language
11	Implement UI switch using Swift language
12	Implement Stepper using Swift language
13	Implement a simple data picker
14	Implement UI elements available in iOS Programming.

Learning Resources:

Reference Books:

1. Professional iOS Programming with Swift by Michael Dippery, Wiley, ISBN: 978-8126556854
2. Swift iOS Programming: 24-Hour Trainer by Abhishek Mishra, Wiley, ISBN: 978-8126559886
3. Programming iOS 11: Dive Deep into Views, View Controllers, and Frameworks by Matt Neuberg, O'Reilly, ISBN : 978-1491999226
4. SwiftUI Essentials - iOS 15 Edition: Learn to Develop IOS Apps Using SwiftUI, Swift 5.5 and Xcode 13 by Smyth, Payload Media, Inc, ISBN: 978-1951442439
5. iOS App Development for Dummies by Jesse Feiler, Wiley, ISBN - 978-8126550548

Supplementary Reading:

Web Resources:

Weblinks:

1. <https://www.hackingwithswift.com/quick-start/swiftui>
2. <https://swift.org/getting-started/swiftui/>
3. <https://getstream.io/blog/learn-swiftui/>

MOOCS: Spoken Tutorial, Khan Academy, Apple Certifications

Pedagogy:

1. Practical learning
2. Problem solving assignment.
3. Conceptual and Contextual learning
4. PPT Presentation & Interactive discussion



COURSE STRUCTURE

Course Code	MCA50190				
Course Title	Next Generation Database				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	4				4
Credits	4				4
Assessment Schema Code	TT1				
<u>Prerequisites:</u>					
Knowledge of RDBMS, MySQL, PHP, Python					
<u>Course Objectives:</u>					
1. To study the usage and applications of Object-Oriented database					
2. To acquire knowledge on variety of NoSQL databases					
3. To acquire knowledge of developing web applications using NOSQL.					
<u>Course Outcomes:</u>					
1. Compare and Contrast NoSQL databases with each other and Relational Database Systems					
2. Critically analyse and evaluate variety of NoSQL databases.					
3. Students able to develop web applications using NOSQL variations.					
4. Students will understand when and why to use NOSQL					
<u>Course Contents:</u>					
Unit I: Introduction to NOSQL (10 Sessions)					
Definition of NOSQL, History of NOSQL and Different NOSQL products, advantages and disadvantages of NOSQL , why to use NOSQL over SQL , big data and NoSQL relation					
Unit II: NOSQL Basics with MongoDB (15 Sessions):					
NOSQL Storage Architecture, CRUD operations with MongoDB, Querying, Modifying, Indexing and ordering suitable examples using MongoDB, Simple objects and array objects using JSON, Encode and Decode JSON Objects using Java/PHP/Python					
Unit III: Working with NOSQL (10 Sessions)					
Surveying Database Internals, NOSQL, Web Frameworks and NOSQL, using MySQL as NOSQL					

Unit IV: NOSQL Basics with Cassandra (10 Sessions)

Introduction to Cassandra Query Language, CQL data modelling, Using CQL, CQL reference

Unit V: Developing Web Applications with NOSQL and NoSQL Administration (15 Sessions)

PHP and MongoDB, python and MongoDB, Creating Blog Application with PHP, NOSQL Database Administration

Sample Lab Assignments

Lab Assignments on Next Generation Database	
1	a) List top 10 stations with most outbound trips (Show station name and number of trips) b) List top10 stations with most in bound trips (Show station name and number of trips) c) List top 5 routes with most trips (Show starting station name, ending station name and number of trips) (4) List the hour number(forexample 13 means 1pm-2pm) and number of trips which start 5 hours from the station "B.U.Central" d) List the hour number(for example 13 means 1pm-2pm) and number of trips which end at the station "B.U. Central"
2	Download a zip code dataset at http://media.mongodb.org/zips.json .Use mongo import to import the zip code dataset into Mongo DB. After importing the data, answer the following questions by using aggregation pipelines: (1) Find all the states that have a city called "BOSTON". Find all the states and cities whose names include the string "BOST". Each city has several zip codes. Find the city in each state with the most number of zip codes and rank those cities along with the states using the city populations. Mongo DB can query on spatial information.

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3	Create a database that stores road cars. Cars have a manufacturer, a type. Each car has a maximum performance and a maximum torque value. Do the following: Test Cassandra's replication schema and Consistency models
4	Master Data Management using Neo4j Manage your master data more effectively The world of master data is changing. Data architects and application developers are swapping their relational databases with graph databases to store their master data. This switch enables them to use a data store optimized to discover new insights in existing data, provide a 360-degree view of master data and answer questions about data relationships in real time
5	Shopping Mall case study using Cassandra, where we have many customers ordering items from the mall, we have suppliers who deliver them their ordered items.

Learning Resources:

Reference Books:

1. Dan Sullivan, "NoSQL for Mere Mortals", 1st Edition, Pearson Education, 2015. (ISBN- 13: 978-9332557338)
2. Pramod J. Sadalage, Martin Fowler, "NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence", 1st Edition, Pearson Education, 2012. (ISBN-13:978-8131775691)

Supplementary Reading:

Web Resources:

MOOCS: NPTEL, Coursera

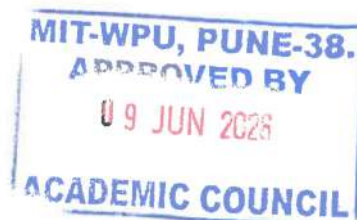
Weblinks:

1. <https://www.tutorialspoint.com/mongodb/index.htm>
2. <https://docs.datastax.com/en/cql/3.3/cql/cqlIntro.html>

Pedagogy:

- Participative learning,
- Discussions,
- Program writing,
- Experiential learning through practical problem-solving,
- Assignment, PowerPoint presentation

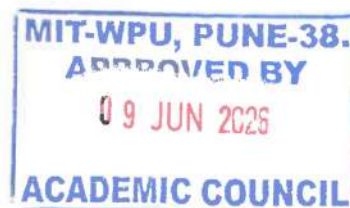
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COURSE STRUCTURE

Course Code	MCA50200				
Course Title	Cloud Computing				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	4				4
Credits	4				4
Assessment Schema Code	TT1				
<u>Prerequisites:</u>					
Knowledge of Operating System, Resource sharing					
<u>Course Objectives:</u>					
1. To provide students with the fundamentals and essentials of Cloud Computing.					
2. To provide students with a sound foundation of Cloud Computing so that they are able to start using and adopting Cloud Computing services and tools in their real-life scenarios.					
3. To enable students to explore some important cloud computing driven commercial systems and applications.					
4. To expose the students to frontier areas of Cloud Computing and information systems, while providing sufficient foundations to enable further study and research.					
<u>Course Outcomes:</u>					
After completion of this course students will be able to:					
1. Explain the core concepts of the cloud computing paradigm: how and why this paradigm shift came about, the characteristics, advantages and challenges brought about by the various models and services in cloud computing.					
2. Apply the fundamental concepts in datacentres to understand the trade-offs in power, efficiency and cost.					
3. Analyse various cloud programming models and apply them to solve problems on the cloud.					
<u>Course Contents:</u>					
Unit 1: Introduction to Cloud computing (10 Sessions)					
Definition of Cloud computing, Introduction to Cloud Computing, Definition of Cloud, Evolution of Cloud Computing, Underlying Principles of Parallel and Distributed Computing, Cloud Characteristics,					

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Elasticity in Cloud, On-demand Provisioning, characteristics of cloud computing, Benefits and limitations of cloud, Cloud Computing vs Grid computing vs Cluster computing, Cloud Types, Understanding cloud architecture.

Unit 2: Cloud Enabling Technologies (15 Sessions)

Service Oriented Architecture REST and Systems of Systems Web Services, Publish, Subscribe Model, Basics of Virtualization, Types of Virtualizations, Implementation Levels of Virtualization, Virtualization Structures, Tools and Mechanisms, Virtualization of CPU, Memory, I/O Devices, Virtualization Support and Disaster Recovery.

Unit 3: Understanding Services and Applications by Type (10 Sessions)

Defining Infrastructure as Service (IaaS), Defining Platform as Service (PaaS), Defining Software as Service (SaaS), Defining Identity as Service (IDaaS), Defining Container as Service (CaaS).

Unit 4: Understanding Abstraction and Virtualization (10 Sessions)

Using Virtualization technologies, Load Balancing and Virtualization, Understanding Hypervisors, Understanding Machine Imaging.

Unit 5: Understanding Cloud Security (7 Sessions)

Cloud Security Fundamentals, Cloud Security Architecture, Cloud Computing Security Challenges.

Unit 6: Understanding Service Oriented Architecture & Service Level Agreements (8 Sessions)

Introducing Service Oriented Architecture, Defining SOA, Communication Managing & Monitoring SOA, Relating SOA & Cloud Computing, Service Level Agreements, Introduction to Map Reduce

Learning Resources:

Reference Books:

1. Cloud Computing bible by Barrie Sosinsky, 1st edition, Wiley Publishing, Inc., 2010
2. Cloud Computing by Toby Velte, Anthony Velte, Robert Elsenpeter, 1st edition, McGraw- Hill, 2009
3. Cloud Security and Privacy by Tim Mather, Subra Kumaraswamy, Shahed Latif, Published by

- O'Reilly, 2009
4. Cloud Computing: Concepts, Technology & Architecture, Erl, Pearson Education, Ricardo Puttini (Author), Zaigham Mahmood, PHI Publication, 1st edition, India, 2014
 5. Cloud Computing by Mehul Mahrishi Kamal Kant Hiran, Ruchi Doshi, Dr. Fagbola Temitayo, 1st edition, 2019

Supplementary Reading:

Web Resources:

MOOCS: Spoken Tutorial, Khan Academy, Swayam Courses

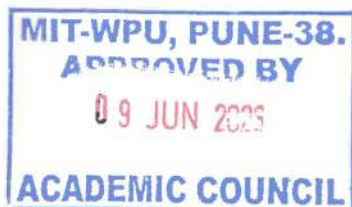
Weblinks:

1. https://www.tutorialspoint.com/cloud_computing/index.htm
2. <https://www.guru99.com/cloud-computing-for-beginners.html>

Pedagogy:

1. Participative learning
2. Discussions
3. Problem solving
4. Assignments
5. Tutorials
6. PPT Presentations
7. Practical Learning

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COURSE STRUCTURE

Course Code	MCA50210				
Course Title	Artificial Intelligence and Machine Learning				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	4				4
Credits	4				4
Assessment Schema Code	TT1				

Prerequisites:

Basics Linear Algebra, Statistics and Probability. Understanding of programming concepts.

Course Objectives:

1. To understand the concepts of artificial intelligence and its applications.
2. To explore various data pre-processing methods
3. To understand the machine learning concepts and its various algorithms.
4. To design, develop and implement AI & ML solutions to a given problem.

Course Outcomes:

Students will be able to:

1. Identify the needs and challenges of Artificial Intelligence for real time applications.
2. Apply various data pre-processing techniques to simplify and speed up machine learning algorithms.
3. To design, develop and implement AI, ML solutions to a given problem.

Course Contents:

Unit 1: Introduction to AI (13 sessions)

The AI Problems – The Underlying Assumptions, what is an AI Technique? Problems: Problems spaces and search, Defining the Problems as a State Space Search. Searching for Solutions - Breadth-first search, Depth-first search, Best First Search, Hill-climbing search, Problem Reduction – Constraint Satisfaction, Crypt Arithmetic, Means – Ends – Analysis, A*, AO* algorithm.

Unit 2: Feature Engineering (10 sessions)

Concept of Feature, preprocessing of data: Normalization and Scaling, Standardization, Managing missing values, Introduction to Dimensionality Reduction, Principal Component Analysis (PCA). Gradient descent

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Feature Extraction: Kernel PCA, Local Binary Pattern. Introduction to various Feature Selection Techniques, Sequential Forward Selection, Sequential Backward Selection.

Unit 3: Machine Learning (10 sessions)

Introduction to Machine Learning, Need for Machine Learning, Machine Learning Process, Types of Machine Learning, Supervised Learning, Unsupervised Learning, Reinforcement Learning, Supervised vs Unsupervised vs Reinforcement Learning, Models of Machine learning: Geometric model, Probabilistic Models, Logical Models, Grouping and grading models, Parametric and non-parametric models

Unit 4: Supervised Learning Algorithms (12 sessions)

Linear Regression, Logistic Regression, Decision Tree, Random Forest, Naive Bayes, K Nearest Neighbour (KNN), Support Vector Machine (SVM).

Evaluation Metrics and Score: MAE, RMSE, R2, Confusion Matrix, Accuracy, Precision, Recall, F1 score, Cross-validation.

Unit 5: Unsupervised Learning Algorithms (15 sessions)

Clustering, Types of Clustering, K-means Clustering, Hierarchical Clustering, C-means, Apriori Algorithm. Evaluation metrics and score: elbow method, extrinsic and intrinsic methods.

List of Experiments:

1. Implement the depth first search algorithm.
2. Implement best first search algorithm.
3. Implement the breadth first search algorithm.
4. Implement linear regression by building basic regression model that will fit a line to the data.
5. Implement the Logistic Regression for classification by using 10-fold cross validation.
6. Implement K-Means clustering/ hierarchical clustering on sales_data_sample.csv dataset (Optional). Determine the number of clusters. Dataset link: <https://www.kaggle.com/datasets/kvanyoga/sample-sales-data>
7. Predict the price of the Uber ride from a given pickup point to the agreed drop-off location.

Perform following tasks:

1. Pre-process the dataset.
2. Identify outliers.
3. Check the correlation.
4. Implement linear regression and random forest regression models.
5. Evaluate the models and compare their respective scores like R2, RMSE, etc.

Dataset link: <https://www.kaggle.com/datasets/yasserh/uber-fares-dataset>

8. Implement K-Nearest Neighbors algorithm on diabetes.csv dataset. Compute confusion matrix, accuracy, error rate, precision and recall on the given dataset (Optional). Dataset link: <https://www.kaggle.com/datasets/abdallamahgoub/diabetes>

Learning Resources:

Reference Books:

1. Artificial Intelligence by Elaine Rich and Kevin Knight. Tata McGraw Hill, 2nd Edition,
2. Introduction to Machine Learning by Ethem Alpaydin, PHI 2nd Edition-2013
3. Introduction to Artificial Intelligence and Expert System by Dan Patterson, Prentice Hall of India Pvt. Ltd., 2nd Edition, New Delhi, 1997.
4. Machine Learning: The Art and Science of Algorithms that Make Sense of Data, By Peter Flach Cambridge University Press, Edition 2012.
5. Introduction to Statistical Machine Learning with Applications in R by Hastie, Tibshirani, Friedman, Springer, 2nd Edition-2012
6. Hastie, Trevor, et al., —The elements of statistical learning: data mining, inference, and prediction, Vol. 2. New York: springer, 2009.
7. Trent hauk, —Scikit-learn, Cookbook, Packt Publishing, ISBN: 9781787286382
8. Goodfellow I., Bengio Y. and Courville, — A Deep Learning, MIT Press, 2016

Web Resources:

1. Foundation of Machine Learning: <https://cs.nyu.edu/~mohri/mlbook/>
2. Dive into Deep Learning: <http://d2l.ai/>
3. A brief introduction to machine learning for Engineers: <https://arxiv.org/pdf/1709.02840.pdf>
4. Feature selection: <https://dl.acm.org/doi/pdf/10.5555/944919.944968>
5. Introductory Machine Learning Notes: <http://lcs.mit.edu/courses/ml/1718/MLNotes.pdf>

Weblinks:

1. <https://www.coursera.org/>
2. <https://www.edx.org/>

Pedagogy:

1. Participative learning,
2. Assignments,
3. Tutorial
4. Project Based Learning



COURSE STRUCTURE

Course Code	MCA50220				
Course Title	Advanced Software Development Methodologies				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	4				4
Credits	4				4
Assessment Schema Code	TT1				
<u>Prerequisites:</u>					
Software Engineering, System analysis and design.					
<u>Course Objectives:</u>					
1. To introduce git for software development					
2. To learn the principles and practices associated with each of the agile development methods.					
3. To apply the principles and practices of agile software development on a project of interest and relevance to the student.					
<u>Course Outcomes:</u>					
After completion of this course students will be able to:					
1. Use git for software development and deployment.					
2. Apply a thorough understanding of Agile principles and specific practices.					
3. Judge, craft and evaluate appropriate adaptations to existing practices or processes depending upon analysis of typical problems.					
<u>Course Contents:</u>					
Unit 1: Git & Version Control (7 Sessions)					
Basic Concepts, Environment setup, Life Cycle, Branches & Merging, working with local repository and Remote Repository.					
Unit 2: Agile Methodology (10 Sessions)					

Introduction, software development with agile, traditional model vs agile model, agile methods classification, manifesto and principles, project management, team interactions, ethics in teams, agility in design and testing, documentations, agile drivers, capabilities, and values.

Unit 3: Agile Processes (10 Sessions)

Lean production - SCRUM, Crystal, Feature Driven Development, Adaptive Software Development, and Extreme Programming: Method overview, lifecycle, work products, roles and practices.

Unit 4: Agility and Knowledge Management (15 Sessions)

Agile information systems, decision making, knowledge management, institutional knowledge evolution cycle, development, acquisition, refinement, distribution, deployment, leveraging, knowledge management in software engineering, managing software knowledge, challenges of migrating to agile methodologies, agile knowledge sharing, story- cards and Story-card Maturity Model (SMM).

Unit 5: Agility and Requirements Engineering (7 Sessions)

Impact of agile processes, current practices, variance, requirement engineering, managing unstable requirements, requirements elicitation, abstraction model, requirements management in agile environment, requirements prioritization, requirements modelling and generation, concurrency.

Unit 6: Agility and Quality Assurance (10 Sessions)

Agile Interaction Design and product development, Agile Metrics and Feature Driven Development (FDD), Financial and Production Metrics in FDD, Quality Assurance with Agile approach, Test Driven Development, Pair programming, Global Software Development.

Learning Resources:

Reference Books:

1. Agile Software Development, Principles, Patterns, and Practices by Robert C. Martin, Pearson Education Publication, 1st edition, 2014.
2. Essential Scrum: A Practical Guide to the Most Popular Agile Process by Kenneth S. Rubin, Addison Wesley publication, 1st edition, 2012.

3. Succeeding with Agile: Software Development Using Scrum, Pearson Addison-Wesley Professional publication, 1st edition, 2009
4. Agile Estimating and Planning by Cohn, Mike, 1st edition, Pearson Education publication, 2006.

Supplementary Reading:

Web Resources:

MOOCS: Spoken Tutorial, Khan Academy, Swayam Courses

Weblinks:

1. <https://github.com/topics/agile-development>
2. <https://opensource.com/article/19/8/guide-agile-project-management>

Pedagogy:

1. Participative learning
2. Discussions
3. Problem solving
4. Assignments
5. Tutorials
6. PPT Presentations
7. Practical Learning

COURSE STRUCTURE

Course Code	MCA50230				
Course Title	Data Mining and Data Warehousing				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	4				4
Credits	4				4
Assessment Schema Code	TT1				
<u>Prerequisites:</u>					
Database Management System, Statistics.					
<u>Course Objectives:</u>					
1. To understand the principles of Data warehousing and Data Mining. 2. To be familiar with the Data warehouse architecture and its Implementation. 3. To know the Architecture of a Data Mining system. 4. To understand the various Data preprocessing Methods. 5. To perform classification and prediction of data.					
<u>Course Outcomes:</u>					
After completion of this course students will be able to:					
1. Understand the basic concepts and familiar with the architecture and its implementation 2. Analyse and evaluate performance of algorithms for Association Rules 3. Apply pre-processing techniques for data cleansing 4. Analyse Classification and Clustering algorithms 5. Analyse and apply mining techniques					
<u>Course Contents:</u>					
Unit 1 : Data Warehousing and Business Analysis: (12 Sessions) Data warehousing Components – Building a Data warehouse –Data Warehouse Architecture – DBMS Schemas for Decision Support – Data Extraction, Cleanup, and Transformation Tools –Metadata – reporting – Query tools and Applications – Online Analytical Processing (OLAP) – OLAP and Multidimensional Data Analysis.					

Unit 2 : Data Mining: (12 Sessions) - Data Mining Functionalities – Data Preprocessing – Data Cleaning – Data Integration and Transformation – Data Reduction – Data Discretization and Concept Hierarchy Generation- Architecture of a Typical Data Mining Systems- Classification of Data Mining Systems. Association Rule Mining: - Efficient and Scalable Frequent Item set Mining Methods – Mining Various Kinds of Association Rules – Association Mining to Correlation Analysis – Constraint-Based Association Mining.

Unit 3 : Classification and Prediction: (12 Sessions) - Issues Regarding Classification and Prediction – Classification by Decision Tree Introduction – Bayesian Classification – Rule Based Classification – Classification by Back propagation – Support Vector Machines – Associative Classification – Lazy Learners – Other Classification Methods – Prediction – Accuracy and Error Measures – Evaluating the Accuracy of a Classifier or Predictor – Ensemble Methods – Model Section.

Unit 4 Cluster Analysis: (12 Sessions) - Types of Data in Cluster Analysis – A Categorization of Major Clustering Methods – Partitioning Methods – Hierarchical methods – Density-Based Methods – Grid-Based Methods – Model-Based Clustering Methods – Clustering High-Dimensional Data – Constraint-Based Cluster Analysis – Outlier Analysis.

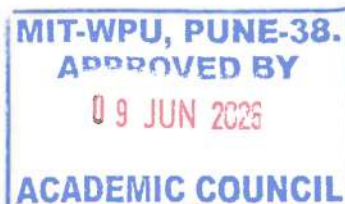
Unit 5: Mining Object, Spatial, Multimedia, Text and Web Data: (12 Sessions)

Multidimensional Analysis and Descriptive Mining of Complex Data Objects – Spatial Data Mining – Multimedia Data Mining – Text Mining – Mining the World Wide Web.

List of Lab Experiments:

1. Creating new Arff File
2. Conversion of Text file into Arff file and vice versa
3. Data Processing Techniques on Data set
4. Association rule mining on data sets
5. Data cube construction – OLAP operations
6. Implementation of Apriori algorithm
7. Implementation of FP- Growth algorithm

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8. Implementation of Decision Tree Induction
9. Calculating Information gains measures
10. Classification of data using Bayesian approach
11. Implementation of K-means algorithm

Learning Resources:

Reference Books:

1. A Jiawei Han, Micheline Kamber and Jian Pei “Data Mining Concepts and Techniques”, Third Edition, Elsevier, 2011.
2. Alex Berson and Stephen J. Smith “Data Warehousing, Data Mining & OLAP”, Tata McGraw – Hill Edition, Tenth Reprint 2007.
3. K.P. Soman, Shyam Diwakar and V. Ajay “Insight into Data mining Theory and Practice”, Easter Economy Edition, Prentice Hall of India, 2006.
4. G. K. Gupta “Introduction to Data Mining with Case Studies”, Easter Economy Edition, Prentice Hall of India, 2006.
5. Pang-Ning Tan, Michael Steinbach and Vipin Kumar “Introduction to Data Mining”, Pearson Education, 2007.

1. Pedagogy:

8. Participative learning
9. Discussions
10. Problem solving
11. Assignments
12. Tutorials
13. PPT Presentations
14. Practical Learning

COURSE STRUCTURE

Course Code	MCA50240				
Course Title	Internet of Things				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	4				4
Credits	4				4
Assessment Schema Code	TT1				
<u>Prerequisites:</u>					
Concepts of hardware, software, computer programming					
<u>Course Objectives:</u>					
1. To understand the fundamentals of Internet of Things. 2. To build a small low cost embedded system using Arduino / Raspberry Pi 3. To apply the concept of Internet of Things in the real world scenario					
<u>Course Outcomes:</u>					
After completion of this course students will be able to:					
1. Attain the fundamental knowledge of Internet of Things 2. To know the sensors used in IoT 3. To understand the IoT communication technologies 4. To build IoT systems using Raspberry Pi, Arduino 5. To learn the case studies on real world IoT systems					
<u>Course Contents:</u>					
Unit 1: Introduction to Internet of Things (12 Sessions)					
Definition of IoT, Key Features and Characteristics of IoT - Sensors, Actuators - Physical Design of IoT – IoT Protocols - IoT communication models, IoT Communication APIs - IoT enabled Technologies – Wireless Sensor Networks, Cloud Computing, Embedded Systems - Domain Specific IoTs – Home, City, Environment, Energy, Agriculture and Industry - IoT vs M2M, Privacy and Security Issues					
Unit 2: Sensors (12 Sessions)					
Light sensor, temperature sensor with thermistor - voltage sensor, ADC and DAC - Temperature and Humidity Sensor DHT11 - Motion Detection Sensors - Wireless Bluetooth Sensors, Level Sensors, USB Sensors - Embedded Sensors, Distance Measurement with ultrasound sensor					
Unit 3: IoT Communication Technologies (12 Sessions)					

2G/GSM - 3G, 4G - GPS, GPRS - WIFI, ZIGBEE, ANT, RFID NFC - Blue tooth, Ethernet, IoT network topology, IoT cloud

Unit 4: IoT Physical Devices and Endpoints (12 Sessions)

Introduction and Architecture of Arduino and Raspberry Pi - Interfaces (serial, SPI, I2C) - Basic Programming

Unit 5: Case Studies and Advanced Topics (12 Sessions)

Various Real time applications of IoT – Case Study on Lighting as a service, Intelligent Traffic systems, Smart Parking, and Smart water management - Connecting IoT to cloud – Cloud Storage for IoT - Data Analytics for IoT – Software & Management Tools for IoT

List of Lab Experiments:

1. Controlling the Light Emitting Diode (LED) with a push button
2. Interfacing the RGB LED with the Arduino
3. Controlling the LED blink rate with the potentiometer interfacing with Arduino
4. Detection of the light using photo resistor
5. Interfacing of temperature sensor LM35 with Arduino
6. Interfacing Servo Motor with the Arduino
7. Interfacing of the Active Buzzer with Arduino.
8. Interfacing of the Relay with Arduino.
9. Building Intrusion Detection System with Arduino and Ultrasonic Sensor
10. Directional Control of the DC motor using Arduino

Reference Books:

1. The internet of things - Sean Dodson and Rob van Kranenburg, 2009
2. Architecting the Internet of Things - Dieter Uckelmann, Mark Harrison, Florian Michahelles (auth.), Dieter Uckelmann, Mark Harrison, Florian Michahelles (eds.), Springer-Verlag Berlin Heidelberg, Year: 2011
3. Getting Started with the Internet of Things: Connecting Sensors and Microcontrollers to the Cloud - Cuno Pfister, O'Reilly Media, Year: 2011
4. The Internet of Things: Key Applications and Protocols - Olivier Hersent, David Boswarthick, Omar Elloumi, Wiley, 2012

5. Designing the Internet of Things - Adrian McEwen, Hakim Cassimally – WILEY- First Edition, 2014
6. Internet of Things – Architecture and Design - Raj Kamal - McGraw Hill Publications, First Edition, 2017
7. Getting Started with Raspberry Pi - Matt Richardson and Shawn Wallace, SPD Publications, Third Edition, 2016

Supplementary Reading

Web Links:

https://www.cse.wustl.edu/~jain/cse570-15/ftp/iot_prot/index.html

MOOC Courses:

https://onlinecourses.nptel.ac.in/noc22_cs53/preview

Pedagogy:

Participative learning, Discussions, Power Point presentation, Project Presentation.

COURSE STRUCTURE

Course Code	MCA40230				
Course Title	Lab on Advanced Java				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours			2		2
Credits			1		1
Assessment Schema Code	PR				

Prerequisites:

Object Oriented Concepts, Basic Java Programming, Database and Web Technology concepts

Course Objectives:

1. Explore advanced topic of Java programming for solving problems
2. Be able to put into use the advanced features of the Java language to build and compile robust enterprise- grade applications
3. Provide a sound foundation to the students on the concepts, precepts and practices, in a field that is of immense concern to the industry and business
4. Enhance knowledge to manipulate and store data

Course Outcomes:

After completion of this course students will be able to:

1. To learn and implement the concepts of advanced java like Servlet, JSP, Spring and Hibernate framework.
2. To understand and apply the concepts of RMI and Java Beans.
3. To build the web application using the JDBC and other related concepts.

Course Contents:

Sample LAB Assignments

1. Write a program create simple registration form in servlet (Parameters to consider Name, Password, Email, Country)
2. Write a program to fetch the Result for a Roll No. using servlets and getting the data from the database having attributes as Name, Roll NO, Result and Grade.
3. Write a program to upload a file on server in servlet.
4. Write a program to download a file from server in servlet.
5. Write a program of sending email from servlet.

6. Write a program create simple registration form in JSP (Parameters to consider Name, Password, Email, Country)
7. Write a program that creates simple Login form in JSP (Parameters to consider Name, Password, Email, Country). Extend it by adding logout.
8. Write a program to upload a file on server in JSP.
9. Write a program to download a file from server in JSP. Write a program to create custom JSP Tag.
10. Write a program using WSDL.
11. Write a program explaining SOAP. Write a program explaining RPC.
12. Write a program implementing creating and calling Web Services.
13. Write a program to create simple Spring.
14. Write a program to implement Spring Dependency Injections using both setter method and constructor method.
15. Write a program to implement Spring Beans.
16. Write a program to implement Spring AOP
17. Write a program to create simple Hibernate.
18. Write a program to demonstrate simple Hibernate – XML Mapping.
19. Write a program to perform different database operations using Hibernate Query Language.

Laboratory Exercises / Practical:

Every student will carry out minimum *Six Practical/exercises* based on the above units and submit the journal, which will be evaluated as part of continuous assessment.

Learning Resources:

Reference Books:

1. Java Complete Reference Patric Naughton, Herbert Schildt, TMH, 11th Edition, 2018
2. Complete Reference- J2EE by Jim Keogh, 1st Edition, TMH, 2017
3. Inside Servlets by Dustine R. Callway, Pearson Education, 1st edition, 2009
4. JDBC, Servlet and JSP, Black Book, Santosh Kumar K. Dremtech publication 2nd Ed. Jan 2016
5. Professional JSP Wrox press 2nd Edition 1-Jan 2001.
6. Spring Persistence with Hibernate, Ahmad Seddighi, Packt Publishing Limited, 1st edition, 2009.
7. Java unleashed by Michael Morrison, Sams.net publication, 1st Edition, 1997
8. Spring and hibernate, Santosh Kumar K., 1st edition, McGraw Hill Education, 2009

Supplementary Reading:

Web Resources:

Weblinks:

1. <https://www.javatpoint.com>
2. <https://www.tutorialspoint.com/asp.net/index.htm>
3. www.W3schools.com
4. <https://www.udemy.com/course/advanced-java-programming/>
5. <https://www.edureka.co/blog/advanced-java-tutorial>

MOOCS: Spoken Tutorial, Khan Academy, Swayam Courses on Advanced Java

Pedagogy:

1. Practical learning
2. Problem solving assignment.
3. Conceptual and Contextual learning
4. PPT Presentation & Interactive discussion

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COURSE STRUCTURE

Course Code	MCA40240				
Course Title	Lab on Data Structures and Algorithms				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours			2		2
Credits			1		1
Assessment Schema Code	PR				
<u>Prerequisites:</u>					
Procedural Languages C, Object Oriented Paradigm using C++, Basic Knowledge about Algorithms					
<u>Course Objectives:</u>					
1. To make the students understand about basic concepts and importance of Data structures.					
2. To make the students familiar with different algorithms like Stack, Queues, Linked Lists, Trees, Graphs etc.					
<u>Course Outcomes:</u>					
After completion of this course students will be able to:					
1. Understand and apply data structures.					
2. Use linear and non-linear data structures like stacks, queues, linked list etc.					
3. Learn understanding of tree, graph data structures.					
<u>Course Contents:</u>					
1. Implementation of Linear and Binary Search.					
2. Implementation of Stack, Queue, Circular Queue for Integer.					
3. Implementation of Infix to Postfix Expression.					
4. Evaluation of Postfix Expression using Stack.					
5. Implementation of Bubble Sort.					
6. Implementation of Selection Sort.					
7. Implementation of Insertion Sort.					
8. Implementation of Radix Sort.					
9. Implementation of Quick sort.					
10. Implementation of Merge sort.					
11. Implementation of Heap Tree using Insert					
12. Implementation of Heap Tree using Heapify/Adjust					
13. Implementation of Heap Sort using Adjust/Heapify					

14. Perform Insert, Display, delete, search, sum operation on Singly Linked List
15. Program to Demonstrate of Dynamic Stack and Queue using Linked List
16. Implementation of Dynamic Queue using Link List
17. Implementation of Doubly Link List, Circular Link List
18. Implementation of Binary search Tree Insertion, Deletion, Search & Recursive Tree Traversal
19. Implementation of BFS, DFS of a given graph.

Laboratory Exercises / Practical:

Every student will carry out minimum *Six Practical*/exercises based on the above units and submit the journal, which will be evaluated as part of continuous assessment.

Learning Resources:

Reference Books:

1. An Introduction to Data Structures with Application by Tremblay, J. & Sorenson, P.G., Tata Mcgraw Hill India, 2nd edition, 2001,
2. Data Structures using C and C++ by Langsam, Y., Augenstein, M. J. & Tenenbaum A.M, 2nd Edition, Pearson Education, 2015
3. Data Structures using C by Balagurusamy, E., 1st Edition, Mcgraw Hill Education, 2013
4. Data Structures and Algorithm Analysis in C by Weiss, M.A., 2nd Edition, Pearson India, 2002
5. Data Structures and Algorithm Analysis in C by Horowitz, E., Sartaj .S. & Mehta, D., X Universities Press, 1st edition, 2008
6. Data Structures & Algorithms in Java by Lafore, R., 2nd Edition, Pearson India, 2008
7. Data Structures and Program Design in C by Kruse, R., Tondo, C.L., Leung B., & Mogalla S, Pearson India, 2nd edition, 2006.

Supplementary Reading:

Web Resources:

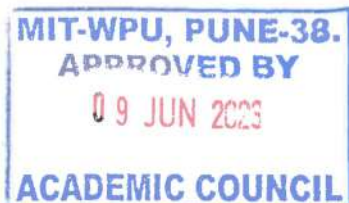
MOOCS: Spoken Tutorial, Khan Academy, Swayam Courses

Weblinks:

1. <https://www.geeksforgeeks.org/best-data-structures-and-algorithms-courses/>
2. <https://plainenglish.io/blog/5-platforms-to-learn-algorithms-and-data-structures-3152238de56c#1-hackerrank>

Pedagogy:

2020





1. Participative learning
2. Discussions
3. Problem solving
4. Assignments

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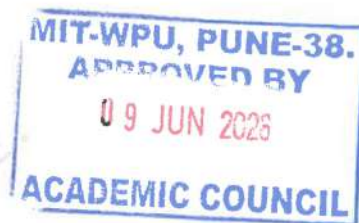
COURSE STRUCTURE

Course Code	MCA40250				
Course Title	Research Paper Writing				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours		2			2
Credits		1			1
Assessment Schema Code	PJ				
<u>Prerequisites:</u>					
Need basics understanding of programming languages and statistics					
<u>Course Objectives:</u>					
1. To Identify the research topic for Research Paper and Research Project Proposal.					
2. To go through the existing literature and identify the gaps.					
3. Writing a Research Paper and Research Project Proposal.					
<u>Course Outcomes:</u>					
After completion of this course students will be able to:					
1. Understand and apply concepts of research methodology.					
2. Design research method for writing research paper and research project proposal.					
3. Writing Research Paper.					
4. Writing Research Project Proposal.					
<u>Course Contents:</u>					
1. Review of research work.					
2. Writing a research paper.					
3. Writing Research Project Proposal.					
<u>Learning Resources:</u>					
1. All the resources in the field of interest include digital library.					
2. Research papers published in various journals and conference proceedings					
<u>Supplementary Reading:</u>					
<u>Web Resources:</u>					

Scopus, IEEE Xplore, Wiley etc.

Pedagogy:

1. Participative learning
2. Conceptual Learning
3. Problem solving
4. Experiential Learning



COURSE STRUCTURE

Course Code	MCA40260				
Course Title	DevOps				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3		-		3
Credits	3		-		3
Assessment Schema Code	TT1				

Prerequisites:

Basic knowledge of Operating Systems, Software Development Life Cycle (SDLC), and Software Development Methodologies.

Course Objectives:

To understand

1. The history of Agile Software Development and DevOps
2. The practices of version control and configuration management, and practices such as Continuous Integration (CI), Continuous Delivery (CD), Continuous Testing, Continuous Deployment, Continuous Monitoring, Continuous Feedback and Continuous Improvement
3. The skills, culture, mindset and tools required to establish a DevOps ecosystem, and the deployment pipeline and its purpose
4. The challenges of provisioning and managing infrastructure, the advantages of automated provisioning and Infrastructure as Code (IaC), and DevOps in a cloud environment

Course Outcomes:

After the completion of the course students will understand and apply their knowledge acquired on,

1. DevOps with reference to SDLC including Agile
2. DevOps practices such as Configuration Management and Version Control, Continuous Integration, Continuous Deployment, and Continuous Testing
3. DevOps in a Cloud environment – Infrastructure Provisioning and the role of Infrastructure as Code (IaC) in automated provisioning and the role of tools/platforms such as Git, Chef, Puppet in establishing a deployment pipeline

Course Contents:



Unit 1: Linux Operating System Basics (11 sessions)

History of Unix and Linux, Difference between Unix and Linux, Difference between Linux and Windows, The structure of Linux File System, Unix/Linux philosophy, Different types of Linux shells, Installing Linux, Basic Linux commands (File operations, Directory operations, Changing file permissions and ownership, Introduction to vi or vim editor), Environment Variables, Redirection, Pipes, Filters, Types of links - soft and hard links, Linux Services, Start and stop services, Introduction to shell programming, Writing simple shell programs, Linux processes, Find and kill a process with id and name, Packages and package management, Different types of packages, Package installation using RPM and YUM, The significance of Linux from DevOps perspective

Unit 2: Introduction to DevOps (10 sessions)

SDLC Models, Challenges in developing software in a team, History of Agile, Agile Manifesto, Agile Principles, Scrum, Extreme Programming, Lean, ITIL, History of DevOps, Purpose of DevOps, DevOps goals, DevOps Stakeholders, DevOps Terminology, Agile and DevOps, ITIL and DevOps, Sample list of DevOps Tools, DevOps practices (Configuration management and version control, CI/CD (Continuous Integration, Continuous Delivery), Continuous Testing, Continuous Deployment, Continuous Monitoring, Continuous Feedback, Continuous Improvement), Role of DevOps Engineer

Unit 3: Git for Version Control (8 sessions)

Introduction to Version Control Systems and Types, Difference between CVCS and DVCS, Introduction to GIT, History of Git, GIT Basics - GIT Command Line, Installing Git- Installing on Linux or Windows, Initial setup, GIT Essentials- Creating repository, Cloning, Check-in and committing, Fetch, Pull and Remote branching, Creating a branch, Switching and merging, Example scenarios for branching and cloning

Unit 4: Chef for Configuration Management (8 sessions)

Overview of Chef, Introduction to Chef Infra, Infrastructure as Code, Chef - basic concepts and terminology (Server, Workstation, Client, Repository, Cookbook, Supermarket, Knife, Resources, Attributes, Data bags, etc.), Chef Architecture, Tools of Chef Infra Platform, Servers and Nodes, Idempotency, Chef Configuration Concepts, Workstation Setup, Organization Setup, Create organization, Test Node Setup, Environments

Unit 5: Puppet for Configuration Management (8 sessions)

Introduction to Puppet, Architecture of Puppet Platform, Benefits of Implementing a Declarative Configuration Tool, Puppet Components and Packages, Open-Source Puppet vs. Puppet Enterprise, Installation and Configuration, Puppet Language Basics, Templates, Installing LAMP with preexisting modules, Installing Apache Tomcat with Puppet Module

Learning Resources:

Reference Books:

1. The DevOps Adoption Playbook: A Guide to Adopting DevOps in a Multi-Speed IT Enterprise by Sanjeev Sharma, IBM press
2. Practical DevOps: Harness the power of DevOps to boost your skill set and make your IT organization perform better by Joakim Verona
3. Linux with Operating System Concepts, Richard Fox, Northern Kentucky University, USA
4. The Phoenix Project: A Novel about It, DevOps, and Helping Your Business Win
5. Beyond the Phoenix Project: The Origins and Evolution of DevOps, Gene Kim (Author, Narrator), John Willis (Author, Narrator), IT Revolution Press (Publisher)
6. Accelerate, the Science of Lean Software and DevOps: Building and scaling High performing Technology organization by Nicole Forsgren, Jez Humble and Gene Kim
7. The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations, Gene Kim, Patrick Debois, John Willis, Jez Humble, IT Revolution Press; Illustrated edition

Web Resources:

1. <https://git-scm.com/doc>
2. https://docs.chef.io/platform_overview/
3. <https://www.puppet.com/why-puppet/about-puppet>
4. <https://www.puppet.com/docs/pdf>

Weblinks:

1. <https://www.coursera.org/>
2. <https://www.atlassian.com/devops>

Pedagogy:

1. Participative Learning
2. Discussions
3. Assignments
4. Tutorials

COURSE STRUCTURE

Course Code	MCA40270				
Course Title	Full Stack Development				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3				3
Credits	3				3
Assessment Schema Code	TT1				
<u>Prerequisites:</u>					
Basic knowledge of computer fundamentals.					
<u>Course Objectives:</u>					
<					

Course Contents:

Unit 1: JavaScript and Modern Syntax (8 Sessions)

Introduction to JavaScript and Execution Environment, Statements, Comments, Variables (var, let, const), Data Types, Type Conversion, Operators, Arrays, Strings, Loops (for, while, for...of), Functions (Function Declarations, Arrow Functions), Objects and Methods, Introduction to ES6+ features: Destructuring, Template Literals, Spread & Rest Operators

Unit 2: DOM Manipulation and Browser Events (9 Sessions)

Introduction to the Document Object Model (DOM), Selecting and Manipulating DOM Elements (querySelector, getElementById), Working with DOM Nodes (Add, Remove, Update Content), Attributes and Class Manipulation, Events and Event Listeners (Click, Mouse, Keyboard, etc.), Event Delegation, Bubbling, Capturing, Virtual DOM vs Real DOM: Conceptual Introduction, Modern Best Practices (Separation of JS & HTML)

Unit 3: React Fundamental and Form Handling (10 Sessions)

Introduction to MERN Stack and Serverless Hello World, Introduction to React and JSX, React Functional Components, Props and Component Composition, Form Handling and Controlled Components, Validation Techniques (Client-side), Children Props and Dynamic Composition, Introduction to React Router for SPA Navigation

Unit 4: React State, APIs and GraphQL (10 Sessions)

React useState, useEffect, and Async State Initialization, Event Handling in React, State vs. Props, Lifting State Up, Stateless vs Stateful Components, Component Hierarchy, RESTful API with Express.js – CRUD and Routing, Introduction to GraphQL – Concepts, Benefits, GraphQL Schema, Field Specification, Queries, Mutations, GraphQL Integration, Query Variables, Input Validation, Error Handling

Unit 5: MongoDB and Full-Stack Integration (8 Sessions)

Introduction to MongoDB: Documents, Collections, Databases, Mongo Shell and MongoDB CRUD Operations (Create, Read, Update, Delete), Aggregation, Projection, MongoDB Node.js Driver, Schema Initialization with Mongoose, Front-End and Back-End Module Organization, Modularization using Webpack and Introduction to Vite (comparison), Hot Module Replacement (HMR), Define Plugin, Build Optimization, Managing Environment Variables with .env and dotenv.

Learning Resources:

Reference Books:

1. JavaScript and JQuery: Interactive Front-End Web Development, Jon Duckett, Wiley India Pvt. Ltd., 2nd Edition, 2023

2. Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Vasani Subramanian, Apress (Springer Nature), 3rd Edition, 2023
3. Eloquent JavaScript: A Modern Introduction to Programming, Marijn Haverbeke, No Starch Press, 4th Edition, 2024
4. Learning React: Modern Patterns for Developing React Apps, Alex Banks & Eve Porcello, O'Reilly Media, 4th Edition, 2024
5. You Don't Know JS Yet: Scope & Closures, Kyle Simpson, Independently Published, 2nd Edition, 2023
6. MongoDB: The Definitive Guide, Shannon Bradshaw & Eoin Brazil, O'Reilly Media, 4th Edition, 2023

Supplementary Reading:

Web Resources:

1. <https://developer.mozilla.org/en-US/docs/Web/JavaScript>
2. <https://react.dev/learn>
3. <https://www.mongodb.com/docs/>
4. <https://webpack.js.org/concepts/>
5. <https://nodejs.org/en/docs>
6. <https://vitejs.dev/>

MOOCs: Online courses for self-learning

1. <https://fullstackopen.com/en/>
2. <https://www.coursera.org/professional-certificates/meta-front-end-developer>
3. <https://www.udemy.com/course/react-redux>

Pedagogy:

1. Concept-Oriented Lectures:
2. Practice-Based Learning
3. Mini-Project Driven
4. Interactive Learning Tools
5. Peer Review & Demo Sessions
6. MOOC/Video Integration

COURSE STRUCTURE

Course Code	MCA40280				
Course Title	Software Testing and Quality Assurance				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3				3
Credits	3				3
Assessment Schema Code	TT1				
<u>Prerequisites:</u>					
Basic Knowledge of software engineering					
<u>Course Objectives:</u>					
1. To study fundamental concepts in software testing 2. To discuss various software testing issues and solutions in software unit test, integration and system testing. 3. To expose the advanced software testing topics, such as object-oriented software testing methods.					
<u>Course Outcomes:</u>					
After completion of this course students will be able to:					
1. To apply white-box testing, black-box testing, and inspection techniques 2. To know how test tools can be used in the testing life cycle and use of testing metrics for product and process. 3. To know the concepts of quality, cost of quality, quality model.					
<u>Course Contents:</u>					
Unit 1: Basic Concepts and Preliminaries [9 Sessions]					
Quality Revolution, Software Quality, Role of Testing, Verification and Validation, Failure, Error, Fault, and Defect, Notion of Software Reliability, Objectives of Testing, What Is a Test Case, Expected Outcome, Concept of Complete Testing, Central Issue in Testing, Testing Activities, Test Levels and Sources of Information for Test Case Studies, White-Box and Black-Box Testing, Test Planning and Design, Monitoring and Measuring Test Execution, Test Tools and Automation, Test Team Organization and Management.					

Unit 2: Testing Techniques - I [9 Sessions]

Unit Testing

Concept of Unit Testing, Static and Dynamic Unit Testing, Mutation Testing, Debugging, Unit Testing Tools: Junit Framework, Control Flow Testing, Basic Idea, Control Flow Graph, Path selection criteria, Generating Test Input, Data Flow Testing, General Idea, Data Flow graphs, Data Flow Terms, Data Flow testing Criteria, Feasible paths and Test selection criteria, Comparison of testing techniques.

Unit 3: Testing Techniques – II [9 Sessions]

Domain Testing, Domain Error, Testing for Domain Errors, Types of Domains Errors, Test selection criteria.

System Integration.

Concept of Integration Testing, System Integration Techniques, Software and Hardware Integration, Test Plan for System Integration.

System Test Categories

Basic Tests, Functionality Tests, Robustness Tests, Interoperability Tests performance Tests, Scalability Tests, Stress Tests, Regression Tests, Documentation Tests, Regulatory Tests.

Unit 4: Software Reliability and Test Team Organization [9 Sessions]

What is Reliability? Definitions of Software Reliability, Factors Influencing Software Reliability. Applications of Software Reliability, Operational Profiles. Reliability Models, Test Team Organization, Test Groups. Software Quality Assurance Group, System Test Team Hierarchy. Effective Staffing of Test Engineers, Recruiting Test Engineers. Retaining Test Engineers, Team Building.

Unit 5: Software Metrics and Quality [9 Sessions]

Measurement during Software Life Cycle Context, Defect Metrics, Metrics for software Maintenance & Requirements, Measurement Principles, Case study for Identifying Appropriate Measures & Metrics for Projects, Five Views of Software Quality, McCall's Quality Factors and Criteria. ISO 9126 Quality Characteristics, ISO 9000:2000 Software Quality Standard.

Selenium Introduction, RC Introduction

Selenium Introduction, what is Selenium and Why Selenium, Use of Automation Testing? Explanation of Selenium and its advantages ∞ Differences between Selenium and QTP, Selenium Components,

Introduction of selenium Components, Selenium Commands, Actions, Asserts, Assessors, Developing Test Cases & Test Suites with Selenium-IDE, Introduction to Selenium RC.

Learning Resources:

Reference Books:

1. Software Testing and Quality Assurance: Theory and Practice by Kshirasagar Naik, Priyadarshi Tripathy, Wiley-Spektrum publication, 1st Edition, 2008.
2. Software Testing and Quality Assurance by Somnath Ashwini A, Tech-Neo Publications LLP, 1st edition, 2019.
3. The Modern Software Testing Handbook: A Quick Guide to Current Software Testing and Quality Assurance Strategies by Anthony Adesanwo, Sleek Press Publication, Kindle edition, 2022.
4. Software Testing: A Craftsman's Approach by Paul C. Jorgensen, Auerbach Publications, 3rd edition, 2008
5. Handbook of Software Quality Assurance, by G. Schulmeyer, Artech House Publishers, 4th edition, 2007
6. Practical Guide to Software Quality Management by J. Horch, Artech House Publishers, 2nd edition, 2003

Supplementary Reading:

Web Resources:

MOOCS: Spoken Tutorial, Khan Academy, Swayam Courses

Weblinks:

1. [https:// http://www.qatutorial.com/](https://http://www.qatutorial.com/)
2. <https://www.softwaretestinghelp.com/best-software-testing-certifications/>

Pedagogy:

1. Discussions
2. Assignments and Tutorials
3. PPT Presentations
4. Practical Learning

COURSE STRUCTURE

Course Code	MCA50250				
Course Title	Deep Learning and Introduction to NLP				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3	1			4
Credits	3	1			4
Assessment Code	TT1				
Prerequisites:					
Machine Learning, Discrete Mathematics, Theory of Computation					
Course Objectives:					
1. To equip students with a comprehensive understanding of deep learning architectures and natural language processing techniques, 2. To understand the application of neural networks such as RNNs and CNNs for language modelling, text processing, and syntactic/semantic analysis, 3. Fostering the ability to solve real-world problems involving unstructured textual data.					
Course Outcomes:					
After successful completion of the course, students will be able to:					
1. Understand the architecture, training, and optimization of deep neural networks including perceptron's, activation functions, and loss functions. 2. Analyse and apply recurrent neural networks (RNN, LSTM, GRU) and attention mechanisms for sequence modelling and generation tasks. 3. Demonstrate foundational knowledge of natural language processing tasks such as morphological, syntactic, and semantic analysis, as well as basic text preprocessing techniques. 4. Construct and evaluate probabilistic and neural language models including n-grams, word embeddings, and contextual representations like BERT. 5. Implement and compare deep learning models using CNNs, RNNs, and hybrid architectures for text classification and syntactic analysis using modern NLP libraries and linguistic resources.					

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Course Contents:

Unit 1- Introduction to neural networks and deep neural networks [12 Sessions]

Introduction to Neural Networks: The Biological Neuron, The Perceptron, Multilayer Feed-Forward Networks, Training Neural Networks: Backpropagation and Forward propagation Activation Functions: Linear, Sigmoid, Tanh, Hard Tanh, Softmax, Rectified Linear, Loss Functions: Loss Function Notation, Loss Functions for Regression, Loss Functions for Classification. Hyperparameters: Learning Rate, Regularization, Momentum, Sparsity, Gradient Descent, vanishing and Exploding gradient.

Unit 2 - Recurrent Architectures and Sequence Modeling [12 Sessions]

Recurrent neural networks (RNNs), LSTM, GRU, bidirectional models, encoder-decoder architectures, attention mechanisms, and sequence generation tasks.

Unit 2 – Introduction to NLP [12 Sessions]

Introduction: Natural Language Processing, Why NLP is hard? Programming languages Vs Natural Languages, Are natural languages regular? Finite automata for NLP challenges and issues in NLP Morphological Analysis, Syntactic Analysis & Semantic Analysis
Basics of text processing: Tokenization, Stemming, Lemmatization, Part of Speech Tagging

Unit 4 - Language Modeling [12 Sessions]

Probabilistic language modeling, Markov models, Generative models of language, Log-Liner Models, Graph-based Models

N-gram models: Simple n-gram models, Estimation parameters and smoothing, Evaluating language models, **Word Embeddings/ Vector Semantics:** Bag-of-words, TFIDF, word2vec, Contextualized representations (BERT)

Unit 5 - CNNs and Recursive Networks in NLP [12 Sessions]

Convolutional neural networks for text classification, pooling strategies, recursive neural networks for syntactic tasks, and hybrid architectures combining CNNs and RNNs, NLP Libraries & Linguistic Resources.

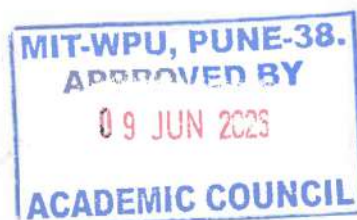
Learning Resources:

Reference Books:

1. Goodfellow, I., Bengio, Y., Courville, A, —Deep Learning, MIT Press, 2016
2. Josh Patterson & Adam Gibson, —Deep Learning
3. Jurafsky, David, and James H. Martin, —Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition, , PEARSON Publication
4. Manning, Christopher D., and Prithvi Schütze “Foundations of Statistical Natural Language Processing, Cambridge, MA: MIT Press

Supplementary Reading:

1. Steven Bird, Ewan Klein, Edward Loper, —Natural Language Processing with Python – Analyzing Text with the Natural Language Toolkit, O'Reilly Publication



2. Richard S. Sutton and Andrew G. Barto, —Reinforcement Learning: An Introduction

Web Links:

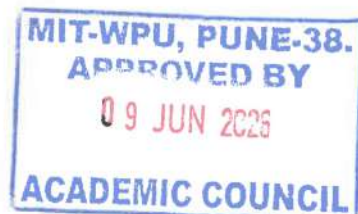
1. https://www.tutorialspoint.com/operating_system/index.htm
2. <https://www.geeksforgeeks.org/operating-systems/>
3. <https://www.studytonight.com/operating-system/>

MOOCs: Online courses for self-learning

Courses by NPTEL and MIT Open Courseware etc

Pedagogy:

- Participative learning
- Discussions
- Presentations
- Videos
- Demonstrations



COURSE STRUCTURE

Course Code	MCA50260				
Course Title	Advanced Internet Technology				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	1		3		4
Credits	1		3		4
Assessment Code	TT1				

Prerequisites:

1. Basic and fundamental concepts of python.

Course Objectives:

1. To design and implement scalable web applications.
2. To apply concepts of databases, forms, and sessions in Django projects.
3. To develop RESTful APIs using Django REST Framework.
4. To deploy Django applications in a production environment.

Course Outcomes:

After completion of this course students will be able to:

1. Create modular, reusable Django applications with best practices.
2. Integrate models, views, and templates to build dynamic websites.
3. Use Django ORM for database operations and admin panel customization.
4. Build REST APIs and consume them using Django.
5. Deploy Django applications on cloud platforms and secure them.

Course Contents:

Unit 1: Django Basics and Project Setup- [10 Sessions]

Django architecture and MVC/MVT design pattern, Virtual environments and project setup, App structure and settings, URL routing and views, Rendering templates with context data.

Unit 2: Models, Forms, and Templates [12 Sessions]

Creating models and migrations, Django ORM: CRUD operations, Admin panel customization, Working with Django forms (ModelForm, regular forms), Template inheritance and filters.

Unit 3: User Authentication and Session Management [14 Sessions]

Built-in user authentication (login, logout, registration), Django messages framework, Sessions and cookies, Permissions and decorators, Custom user models (basic overview).

Unit 4: Django REST Framework [10 Sessions]

Introduction to REST and DRF setup, Serializers and class-based views, CRUD APIs with DRF, API authentication and permissions, Testing APIs using Postman.

Unit 5: Deployment, Security, and Advanced Features [14 Sessions]

Static and media files handling, Django signals, middleware, and pagination, Caching strategies (overview), Deployment on platforms like PythonAnywhere, Heroku, or Render, Basic security practices in Django (CSRF, input validation, HTTPS).

Practical Assignments

1. Set up a basic Django app with templates and routing.
2. Create and manage models using Django ORM.
3. Implement CRUD operations with templates and forms.
4. Customize admin panel for model management.
5. Build a user login and registration system.
6. Develop a blog or e-commerce module with authentication.
7. Create and test REST APIs using Django REST Framework.
8. Integrate frontend with REST APIs (AJAX or simple JS fetch).
9. Deploy a Django project to a cloud hosting platform.
10. Final project: A full-stack web application (e.g., Result Portal, Bookstore, or Inventory Management System).

Learning Resources:

Reference Books:

1. Build a Website with Django 3 – Nigel George, Publisher: Learndjango.com
2. Django 4 By Example – Antonio Mele, Publisher: Packt Publishing
3. Django for Beginners, Django for APIs – William S. Vincent Publisher: WelcomeToCode

Weblinks:

1. Official Django Documentation
<https://docs.djangoproject.com>
2. Django REST Framework Docs
<https://www.django-rest-framework.org>
3. Real Python Django Tutorials
<https://realpython.com/tutorials/django/>
4. DjangoGirls Tutorial
<https://tutorial.djangogirls.org/>
5. Simple is Better Than Complex Blog
<https://simpleisbetterthancomplex.com>

Pedagogy:

- Blended Learning
- Participative learning

- Creative Assignment
- Discussions
- Demonstrations
- Practical Assignment
- Videos

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COURSE STRUCTURE

Course Code	MCA50270				
Course Title	Data Engineering				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3	1			4
Credits	3	1			4
Assessment Code	TT1				
Prerequisites: Basic knowledge of programming (Python/SQL) and databases & Familiarity with data structures and algorithms is beneficial					
Course Objectives: 1. To understand the foundations and ecosystem of data engineering. 2. To develop skills in data ingestion, transformation, and storage. 3. To utilize data pipeline tools and cloud technologies. 4. To ensure data quality, security, and scalability in real-world applications.					
Course Outcomes: After successful completion of the course, students will be able to: 1. Explain the data engineering lifecycle and key components of the modern data ecosystem. 2. Apply appropriate storage and processing systems for handling large-scale structured and unstructured data. 3. Design and orchestrate scalable data pipelines using ingestion, transformation, and scheduling tools. 4. Implement data quality, governance, and monitoring practices in data engineering workflows.					
Course Contents: Unit 1- Introduction to Data Engineering and Ecosystem [15 Sessions] Overview of Data Engineering: Role, responsibilities, and lifecycle, Data types: Structured, semi structured, and unstructured data, Data formats: CSV, JSON, Parquet, Avro, Introduction to Big Data and its characteristics, Overview of the Data Engineering ecosystem: Hadoop, Spark, Kafka, Airflow					

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Unit 2 - Data Storage and Processing [15 Sessions]

Relational Databases and SQL (MySQL/PostgreSQL), NoSQL Databases (MongoDB, Cassandra), Data Lakes and Warehouses (Amazon S3, Delta Lake, Snowflake), Batch vs Stream Processing, Distributed computing with Apache Spark: RDDs, Data Frames, and Transformations.

Unit 3 – Data Ingestion and Transformation [15 Sessions]

Data ingestion tools: Apache Kafka, Sqoop, Flume, ETL vs ELT processes, Orchestrating data pipelines using Apache Airflow, Data transformation techniques using Python and SQL, Schema evolution and change data capture (CDC).

Unit 4 - Data Quality, Governance, and Monitoring [15 Sessions]

Data quality dimensions and validation, Version control for data and pipelines, Data lineage and cataloging tools (e.g., Apache Atlas, Amundsen), Monitoring pipelines: Logs, alerts, and dashboards (e.g., Prometheus, Grafana), Security and access control in data platforms

Learning Resources:

Reference Books:

1. “Designing Data-Intensive Applications” by Martin Kleppmann
2. “Fundamentals of Data Engineering” by Joe Reis and Matt Housley
3. “Streaming Systems” by Tyler Akidau et al.
4. “Data Engineering with Python” by Paul Crickard

Supplementary Reading:

1. Documentation from Spark, Kafka, Airflow, Snowflake
2. Blogs: Towards Data Science, Data Engineering Weekly

Web Links:

1. https://www.tutorialspoint.com/operating_system/index.htm
2. <https://www.geeksforgeeks.org/operating-systems/>
3. <https://www.studytonight.com/operating-system/>

MOOCs: Online courses for self-learning

Courses by NPTEL and MIT Open Courseware etc

Pedagogy:

- Participative learning
- Discussions



- Presentations
- Videos
- Demonstrations

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COURSE STRUCTURE

Course Code	MCA50280				
Course Title	Information Security Audit				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3	1			4
Credits	3	1			4
Assessment Schema Code	TT1				

Prerequisites:

Computer Networks, Network Security

Course Objectives:

1. Explore system security related incidents and gain insight on potential defenses and counter measures against common threat/vulnerabilities.
2. Install, configure and troubleshoot information security devices.
3. Gain experience using tools and common processes in information security audits and analysis of compromised systems.

Course Outcomes:

After completion of the course the student should be able to:

1. Contribute to managing information security, Audit and Co-ordinate responses to information security incidents
2. Install and configure information security devices
3. Support teams to prepare for and undergo information security audits and manage their work to meet requirements
4. Work effectively with colleagues
5. Maintain a healthy, safe and secure working environment and provide data/information in standard formats.

Course Contents:

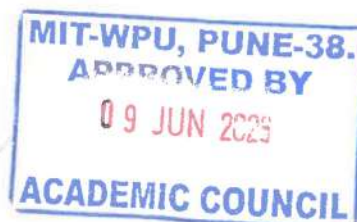
Unit I: Information Security Fundamental (7 Sessions)

Definitions & challenges of security, Attacks & services, Security policies, Security Controls, Access control structures, Cryptography, Deception, Ethical Hacking, Firewalls, Identify and Access Management (IdAM).

Unit II: System Security (5 Sessions)

System Vulnerabilities, Network Security Systems, System Security, System Security Tools, Web Security, Application Security, Intrusion Detection Systems.

Unit III: Information Security Management (5 Sessions)



Monitor systems and apply controls, security assessment using automated tools, backups of security devices, Performance Analysis, Root cause analysis and Resolution, Information Security Policies, Procedures, Standards and Guidelines

Unit IV: Incident Management (5 Sessions)

Security requirements, Risk Management, Risk Assessment, Security incident management, third party security management, Incident Components, Roles.

Unit V: Incident Response (7 Sessions)

Incident Response Lifecycle, Record, classify and prioritize information security incidents using standard templates and tools, Responses to information security incidents, Vulnerability Assessment, Incident Analysis.

Unit VI: Conducting Security Audits (10 Sessions)

Common issues in audit tasks and how to deal with these, Different systems and structures that may need information security audits and how they operate, including: servers and storage devices, infrastructure and networks , application hosting and content management, communication routes such as messaging, Features, configuration and specifications of information security systems and devices and associated processes and architecture, Common audit techniques, Record and report audit tasks, Methods and techniques for testing compliance.

Unit VII: Information Security Audit Preparation (12 Sessions)

Establish the nature and scope of information security audits, Roles and responsibilities, Identify the procedures/guidelines/checklists, Identify the requirements of information security, audits and prepare for audits in advance, Liaise with appropriate people to gather data/information required for information security audits.

Unit VIII: Self and Work Management (9 Sessions)

Establish and agree work requirements with appropriate people, Keep the immediate work area clean and tidy, utilize time effectively, Use resources correctly and efficiently, Treat confidential information correctly, Work in line with organizations policies and procedures, Work within the limits of their job role

Learning Resources:

Reference Books:

1. William Stallings, Lawrie Brown, Computer Security: Principles and Practice, 3rd edition, 2014.
2. Nina Godbole, Information Systems Security: Security Management, Metrics, Frameworks and Best Practices, Wiley, 2017
3. Nina Godbole, Sunit Belapure, Cyber Security- Understanding cyber-crimes, computer forensics and legal perspectives, Wiley Publications, 2016

4. Andrew Vladimirov Michajlowski, Konstantin, Andrew A. Vladimirov, Konstantin V. Gavrilenko, 4. Assessing Information Security: Strategies, Tactics, Logic and Framework, IT Governance Ltd, O'Reilly, 2010
5. Charles P. Pfleeger, Security in Computing, 4th Edition, Pearson, 2009.
6. Christopher J. Alberts, Audrey J. Dorofee, Managing Information Security Risks, Addison-Wesley Professional, 2004
7. Peter Zor, The Art of Computer Virus Research and Defense, Pearson Education Ltd, 2005
8. Lee Allen, Kevin Cardwell, Advanced Penetration Testing for Highly-Secured Environments - Second Edition, PACKT Publishers, 2016
9. Chuck Easttom, System Forensics Investigation and Response, Second Edition, Jones & Bartlett 5. Learning, 2014

Supplementary Reading:

Web Resources:

MOOCS: Coursera

Weblinks:

1. <https://www.auditboard.com/blog/what-is-security-audit/>
2. <https://www.techtarget.com/searchcio/definition/security-audit>
3. <https://www.isaca.org/resources/isaca-journal/issues/2016/volume-5/information-systems-security-audit-an-ontological-framework>

Pedagogy:

- Participative learning,
- discussions,
- problem solving,
- assignments,
- Tutorials

COURSE STRUCTURE

Course Code	MCA50290				
Course Title	Computer Vision & Digital Image Processing				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3	1			4
Credits	3	1			4
Assessment Schema Code	TT1				
<u>Prerequisites:</u> Basic understanding of signal processing					
<u>Course Objectives:</u>					
1. To become familiar with digital image fundamentals 2. To articulate simple image enhancement techniques in Spatial and Frequency domain. 3. To learn concepts of degradation function, restoration and segmentation techniques. 4. To explore various feature extraction methods and dimensionality reduction techniques for use in image analysis and classification tasks. 5. To familiarize students with modern object recognition, motion estimation.					
<u>Course Outcomes:</u>					
After completion of this course students will be able to:					
1. explain the fundamental steps of a digital image processing system and perform basic pixel-level operations. 2. apply spatial and frequency domain techniques to enhance the visual quality of images. 3. implement image restoration and segmentation techniques including morphological operations and thresholding. 4. extract and analyze image features for image comparison and recognition. 5. utilize object recognition and motion estimation algorithms to solve real-world problems in computer vision applications.					
<u>Course Contents:</u>					
Unit 1: Introduction to Digital Image Processing: [14 Sessions]					
Digital image processing, Applications of digital image processing, Fundamental steps in digital image processing, and Components of an image processing system. Digital image fundamentals: Image sampling and quantization, some basic relationships between pixels, Linear and nonlinear operation, Colour Models					

Unit 2: Image enhancement in spatial and frequency domain: [12 Sessions]

Image enhancement in spatial domain: Some basic gray level transformations, Histogram processing, Enhancement using arithmetic/logic operations, Basics of spatial filtering, smoothing spatial filters, Sharpening spatial filters

Image enhancement in the frequency domain: Introduction to the Fourier transform and the frequency domain, Smoothing frequency domain filters, Sharpening frequency domain filters, homomorphic filtering.

Unit 3: Image restoration and Segmentation: [12 sessions]

Image Restoration: A model of the image degradation/restoration process, Noise models, Restoration in the presence of noise only-spatial filtering, Periodic noise reduction by frequency domain filtering.

Morphological image processing and image segmentation: Preliminaries, Dilation and erosion, Opening and closing, the hit-or-miss transformation, Some basic morphological algorithms, Detection of discontinuities, Edge linking and boundary detection, Thresholding, Region-based segmentation, Segmentation by morphological watersheds.

Unit 4: Feature Extraction: [10 Sessions]

Importance of Features, Feature extraction techniques, Histogram of Oriented Gradient (HOG), Scale Invariant Feature Transform (SIFT), Background subtraction techniques, Image Matching, Principal Component Analysis (PCA)

Unit 5: Object Recognition and Motion Estimation: [12 sessions]

Object Recognition techniques: Viola-Jones, Yolo, Deep learning algorithms for Object Recognition. Optical Flow, Gaussian Mixture Model (GMM), Structure of Motion, Motion Estimation.

List of Tutorials:

1. Read and Display an Image, Conversion between colour, gray-scale and black & white image
Negative of an Image (Binary & Gray Scale)
2. Image arithmetic operations
3. Image logical operations
4. Implementation of Relationships between Pixels
5. Image sampling and quantization
6. Analysis of spatial and intensity resolution of images
7. Intensity transformation of images
8. Histogram calculation and equalization
9. Implement Histogram of Oriented Gradient (HOG) for Feature extraction.
10. Write a Program to apply Scale Invariant Feature Transform on input image.
11. Implement frame differencing technique for background subtraction from video.

Learning Resources:

Reference Books:

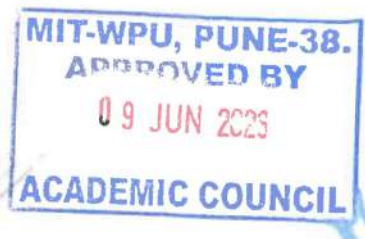
1. Rafael C. Gonzales, Richard E. Woods, "Digital Image Processing", Third Edition, Pearson Education, 2010.
2. Anil Jain K. "Fundamentals of Digital Image Processing", PHI Learning Pvt. Ltd., 2011.
3. Computer Vision - A modern approach, by D. Forsyth and J. Ponce, Prentice Hall Robot Vision, by B. K. P. Horn, McGraw-Hill.
4. Introductory Techniques for 3D Computer Vision, by E. Trucco and A. Verri, Publisher: Prentice Hall.
5. Computer Vision, D. H. Ballard, C. M. Brown, Prentice-Hall, Englewood Cliffs, 1982.
6. Image Processing, Analysis and Machine Vision: Milan Sonka, Vaclav Hlavac, Roger Boyle (Thomson Brooks / Cole Edition).
7. William K Pratt, "Digital Image Processing", John Willey, 2002.
8. Malay K. Pakhira, "Digital Image Processing and Pattern Recognition", First Edition, PHI Learning Pvt. Ltd., 2011.

Supplementary Reading:

MOOCS: Coursera, Udemy, Udacity, Simplilearn, Edx.

Pedagogy:

- Participative learning,
- Discussions, algorithm, demo using latest tools,
- Experiential learning through practical problem solving, assignment,
- Power Point presentations and Case Study



COURSE STRUCTURE

Course Code	MCA50300				
Course Title	ReactJS				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3	1			3
Credits	3	1			4
Assessment Schema Code	TT1				

Prerequisites:

Object Oriented Concepts, Basic Java Programming, Database and Web Technology concepts

Course Objectives:

1. Understanding React Fundamentals
2. Building Single Page Applications (SPAs)
3. Component-based Architecture
4. Handling Data and APIs through Forms and User Input

Course Outcomes:

After completion of this course students will be able to:

1. To build user interfaces specifically for single-page applications.
2. To enable fast and quality assured application development that in turn saves Time for both - clients and developers.
3. To create reusable UI components and hence reduce the code.

Course Contents:

Unit 1: Introduction to ReactJS (12 Sessions)

What is ReactJS?, History of ReactJS., Why to learn ReactJS?, Features of ReactJS, Advantages of ReactJS, Aspects of ReactJS.

Unit 2: Networking with Java (12 Sessions)

Setting up Environment for ReactJS, NodeJS Installation, Downloading and adding ReactJS Libraries, creating your first ReactJS project, Use of create-react-app, ReactJS Project Structure

Unit 3: ReactJS ES6 (12 Sessions)

Classes, Arrow Functions, Variables, Array Methods, DeStructuring, Modules, Ternary Operators and Spread Operators.

Unit 4: React JS Fundamentals (12 Sessions)



React JSX, React Component, Function Based Component, Class Based Component, React Component API, Component Lifecycle, React State, React Props, React State vs React Props

Unit 5: Form Handling in ReactJS (12 Sessions)

React Forms, React Events, React Lists, React Keys, React Refs, React Fragments

Laboratory Exercises / Practical:

Every student will carry out minimum *Five Practical*/exercises based on the above units and submit the journal, which will be evaluated as part of continuous assessment.

Learning Resources:

Reference Books:

1. "FullStack React" The Complete Guide to ReactJS and Friends
2. The Road to Learn React: Your journey to master plain yet pragmatic React.js by Robin Wieruch
3. React in Action by Mark Tielens Thomas
4. Learning React, 2nd Edition.

Supplementary Reading:

Web Resources:

Weblinks:

1. <https://www.javatpoint.com>
2. <https://www.tutorialspoint.com>
3. www.W3schools.com
4. <https://www.udemy.com>

MOOCS: Spoken Tutorial, Khan Academy, Swayam Courses on React Js

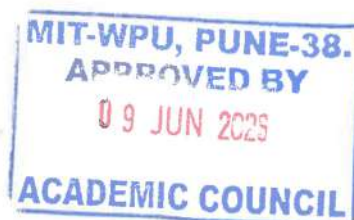
Pedagogy:

1. Practical learning
2. Problem solving assignment.
3. Conceptual and Contextual learning
4. PPT Presentation & Interactive discussion

COURSE STRUCTURE

Course Code	MCA50310				
Course Title	Big Data Analytics				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3	1			4
Credits	3	1			4
Assessment Scheme Code	TT1				
<u>Prerequisites:</u>					
Basics of Operating System, Algorithms, Databases					
<u>Course Objectives:</u>					
1. To teach the fundamentals of Big Data Analytics 2. To teach the architecture of distributed Operating system for Hadoop 3. To teach Hadoop framework and components 4. To teach Hadoop ecosystem tools.					
<u>Course Outcomes:</u>					
After completion of this course students will be able to:					
1. The students will learn Hadoop history, ecosystem, important features of Big Data and its architecture. 2. The students will learn distributed systems, different components of Hadoop ecosystem and Apache Hadoop 3. The students will be able to apply Big Data concepts in Real World.					
<u>Course Contents:</u>					
Unit 1: Introduction: [10 Sessions] What is Big Data? Different types of data generated. Examples of giant companies generating big data, Applications of Big data, what is Data Science?					
Unit 2: Characteristics of Big Data: [12 Sessions] Volume, Velocity and Variety, Other Vs like variability, veracity, valence and value, Big Data Analytics, Case Studies related to Big Data Architecture					
Unit 3: Introduction to Hadoop: (14 Sessions) Distributed File System, Google File System, History of Hadoop, Concept of commodity server, cluster, Hadoop Vs RDBMS, Data integrity in Hadoop, Basic overview of Hadoop Installation, Hadoop Shell Commands					

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Unit 4: Hadoop Framework: (14 Sessions)

HDFS: - Block Size, Architecture - Namenode, Datanode, Secondary namenode, Federation, Anatomy of File Read and Write, MapReduce: MapReduce programming model, Mapper and Reducer, Example of a map reduce job, Matrix multiplication using MapReduce.YARN Architecture: Resource Manager, Node Manager, Application Master, Container, Anatomy of MapReduce Job run in YARN.

Unit 5: Hadoop Ecosystem: (10 Sessions)

Overview and comparison of different ecosystem tools like Apache Flume, Pig, Scoop, Hive, HBase, Storm, Zookeeper, Oozie

Learning Resources:

Reference Books:

1. Hadoop, The Definitive Guide, Tom White, 4th edition, O'Reilly
2. Data Analytics with Hadoop, by Benjamin Bengfort and Jenny Kim

Supplementary Reading:

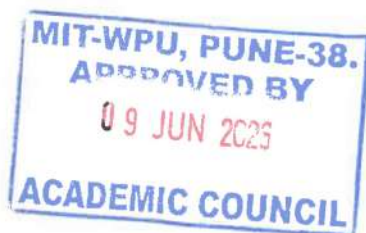
MOOCS: Coursera, Udemy, Udacity, Simplilearn, Edx

Web Resources:

Weblinks: <https://www.tutorialspoint.com>

Pedagogy:

1. Participative learning
2. Discussions
3. Algorithm
4. Flowchart & program writing
5. Demonstrations
6. Practical Assignment
7. Videos
8. Demonstrations



COURSE STRUCTURE

Course Code	MCA50320				
Course Title	Data Privacy and Security				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3	1			4
Credits	3	1			4
Assessment Schema Code	TT1				

Prerequisites:

Basic knowledge of computer networks, cryptography, and information systems.

Course Objectives:

6. To introduce core concepts and importance of data privacy and protection.
7. To understand security threats, risk analysis, and data breach prevention.
8. To learn compliance frameworks, privacy regulations, and their implementations.
9. To explore technologies enabling privacy, such as encryption, anonymization, and secure computation.

Course Outcomes:

After completion of this course students will be able to:

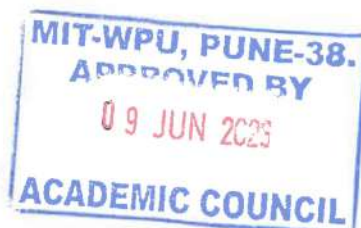
6. Demonstrate understanding of data privacy principles and security mechanisms.
7. Analyze risks and apply techniques for securing personal and organizational data.
8. Implement privacy-preserving technologies using real-world applications.
9. Evaluate data protection laws and compliance strategies such as GDPR, HIPAA, etc.

Course Contents:

Unit 1: Introduction to Data Privacy and Security [12 Sessions]

Concept of Privacy, Definitions and Importance, Data Protection Principles, Types of Data and Classification, Threat Landscape, Data Breaches, Challenges in Data Privacy.

Unit 2: Privacy and Data Protection Regulations [12 Sessions]



Overview of GDPR, HIPAA, CCPA, Indian Personal Data Protection Bill, Compliance Requirements, Consent Management, Cross-border Data Transfers, Roles and Responsibilities under Regulatory Frameworks.

Unit 3: Risk Analysis and Data Governance [12 Sessions]

Privacy Risk Assessment, Threat Modeling, Risk Management Lifecycle, Privacy Impact Assessments (PIAs), Data Governance Policies, Data Minimization, Retention and Access Control Policies.

Unit 4: Security Techniques for Privacy Protection [12 Sessions]

Data Masking, Tokenization, Encryption Techniques, Homomorphic Encryption, Secure Multi-party Computation, Anonymization and De-identification Methods, Privacy in Cloud and Mobile Environments.

Unit 5: Emerging Trends and Technologies [12 Sessions]

Privacy by Design and by Default, AI and Data Privacy, Blockchain and Data Integrity, Privacy-Enhancing Technologies (PETs), Secure APIs, Cybersecurity Insurance, Future Directions in Privacy Engineering.

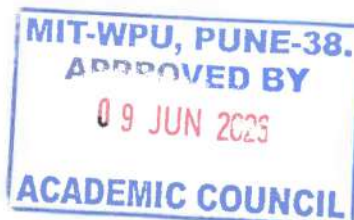
Learning Resources:

Reference Books:

1. Data Privacy and GDPR Handbook, Sanjay Sharma, Wiley India, 1st Edition, 2020
2. Privacy, Big Data, and the Public Good: Frameworks for Engagement, Julia Lane et al., Cambridge University Press, 1st Edition, 2014
3. Understanding Privacy, Daniel J. Solove, Harvard University Press, 1st Edition, 2008
4. Security and Privacy in the Digital Era, John R. Vacca, Elsevier, 1st Edition, 2016

Supplementary Reading:

Web Resources:



1. <https://cloudian.com/guides/data-protection/data-protection-and-privacy-7-ways-to-protect-user-data/>
2. <https://www.geeksforgeeks.org/difference-between-data-privacy-and-data-protection/>

MOOCs: Online courses for self-learning

1. <https://www.udemy.com/course/fundamentals-of-data-privacy/?couponCode=ST8MT220425G3>
2. <https://www.udemy.com/course/certified-data-privacy-and-protection-auditor/?couponCode=ST8MT220425G3>

Pedagogy:

1. Conceptual Lectures supported with real-life use cases.
2. Regulation Mapping Exercises using real policies and compliance tools.
3. Lab Assignments on anonymization, encryption, and secure data sharing.
4. Case Studies: Analyzing major data breach incidents and responses.
5. Group Discussions and Presentations on global privacy laws.
6. Integration of MOOCs and open-source tools for practical exposure

COURSE STRUCTURE

Course Code	MCA40290				
Course Title	Lab on Full Stack Development				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours			2		2
Credits			1		1
Assessment Schema Code	PR				
<u>Prerequisites:</u>					
Basic knowledge of computer fundamentals.					
<u>Course Objectives:</u>					
1. To utilise React JS to build a front-end User Interface. 2. To understand the usage of API to create a web application using Express JS. 3. To store and model data in a NO-SQL Database.					
<u>Course Outcomes:</u>					
After completion of this course students will be able to:					
1. Manipulate and interact with the DOM using JavaScript and handle various browser events. 2. Build dynamic, stateful user interfaces using React and implement RESTful and GraphQL APIs using Express. 3. Develop full-stack web applications using MongoDB, Node.js, React, and Express, with knowledge of module bundling and deployment practices using Webpack.					
<u>Course Contents:</u>					
Topic wise lab assignments					
1. Assignment 1: JavaScript Calculator using Functions a. Create a calculator using JavaScript functions to perform basic arithmetic operations (addition, subtraction, multiplication, division). Use if-else or switch statements.					

2. Assignment 2: DOM-based To-Do List

- a. Build an interactive to-do list using HTML, CSS, and JavaScript. Use DOM methods to:
- b. Add tasks, Mark them as completed, Delete tasks dynamically.

3. Assignment 3: JavaScript Form Validation

- a. Create a user registration form with fields like name, email, phone, and password. Validate inputs using JavaScript and show appropriate error messages.

4. Assignment 4: React Profile Card using Props

- a. Set up a React app and create a component to display a user's profile card. Use props to pass name, image URL, and short description.

5. Assignment 5: Controlled React Form

- a. Design a controlled form in React that accepts user inputs and displays the entered data in real-time below the form.

6. Assignment 6: React Counter App using useState

- a. Create a counter application in React with buttons to increment, decrement, and reset the value. Use React Hooks (useState).

7. Assignment 7: REST API using Express.js

- a. Create a basic RESTful API using Node.js and Express.js with routes to:
- b. Get all users, Add a user, Update user by ID, Delete user by ID.

8. Assignment 8: MongoDB Integration using Mongoose

- a. Design a schema for a "Product" model using Mongoose. Implement routes in Express to perform CRUD operations on MongoDB.

9. Assignment 9: Full-Stack Integration (React + Express + MongoDB)

Create a complete MERN app:

Frontend: React form to add and display users.

Backend: Express routes to interact with MongoDB.

Use Axios to connect frontend to backend.

Learning Resources:

Reference Books:

1. JavaScript and JQuery: Interactive Front-End Web Development, Jon Duckett, Wiley India Pvt. Ltd., 2nd Edition, 2023
2. Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Vasani Subramanian, Apress (Springer Nature), 3rd Edition, 2023
3. Eloquent JavaScript: A Modern Introduction to Programming, Marijn Haverbeke, No Starch Press, 4th Edition, 2024
4. Learning React: Modern Patterns for Developing React Apps, Alex Banks & Eve Porcello, O'Reilly Media, 4th Edition, 2024
5. MongoDB: The Definitive Guide, Shannon Bradshaw & Eoin Brazil, O'Reilly Media, 4th Edition, 2023

Supplementary Reading:

Web Resources:

1. <https://developer.mozilla.org/en-US/docs/Web/JavaScript>
2. <https://react.dev/learn>
3. <https://expressjs.com/>
4. <https://nodejs.org/en/docs>
5. <https://vitejs.dev/>

MOOCs: Online courses for self-learning

1. <https://www.udemy.com/course/the-complete-javascript-course>
2. <https://fullstackopen.com/en/>
3. <https://www.coursera.org/professional-certificates/meta-front-end-developer>
4. <https://www.udemy.com/course/react-redux>

Pedagogy:

1. Mini-Project Driven
2. Interactive Learning Tools
3. Peer Review & Demo Sessions
4. MOOC/Video Integration

COURSE STRUCTURE

Course Code	MCA40300				
Course Title	Mini Project				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours				3	3
Credits				1	1
Assessment Scheme Code	PJ				

Prerequisites:

Technologies like Python and Artificial Intelligence and Machine Learning

Course Objectives:

Course Objective: -

Knowledge:

To understand concepts with respect to

- i. Basic knowledge of Artificial Intelligence and Machine Learning, Concept
- ii. Basics of Python Programming

Skills:

Design skills to

- i. Identify and develop various Software Applications by using AIML and Python Programming

Attitude:

To develop following

- i) Understanding concepts of application development using AIML and Python Programming technologies available.

Course Outcomes:

1. Students will be able to design and create Software applications using AIML and Data Science using Python programming.
2. Students will do so by leveraging the Python programming language and developing the models.
3. Hands on training on various algorithms of machine learning.

Course Contents:

Syllabus Mini Project II

Course Contents:

1	INTRODUCTION
1.1	EXISTING SYSTEM, PROPOSED SYSTEM AND NEED AND OBJECTIVE OF SYSTEM
1.2	SCOPE OF WORK
1.3	OPERATING ENVIRONMENT – HARDWARE AND SOFTWARE
1.4	DESCRIPTION OF TECHNOLOGY USED
2	PROPOSED SYSTEM
2.1	FEASIBILITY STUDY
2.2	USER REQUIREMENTS
3	ANALYSIS AND DESIGN
3.1	OBJECT DIAGRAM
3.2	CLASS DIAGRAM
3.3	USE CASE DIAGRAM
3.4	SEQUENCE DIAGRAM
3.5	ACTIVITY DIAGRAM
3.6	MODULE HIERARCHY DIAGRAM
3.7	USER INTERFACE DESIGN (SCREENS ETC.)
3.8	TABLE SPECIFICATIONS
3.9	TEST PROCEDURES AND IMPLEMENTATION
4	LIMITATIONS
5	FUTURE SCOPE
6	CONCLUSION
7	REFERENCES

MIT-WPU, PUNE-38.
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COURSE STRUCTURE

Course Code	DCA50010				
Course Title	Internship / Capstone Project				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours			45		45
Credits			15		15
Assessment Schema Code	PR				

Prerequisites: Live Application in Industry.

Course Objectives:

1. To enhance students' knowledge in a particular technology and to Increase self- confidence of students and helps in finding their own proficiency.
2. To provide comprehensive learning platform to students where they can enhance their employ ability skills and become job ready along with real corporate exposure.
3. To cultivate student's leadership ability and responsibility to perform or execute the given task.

Course Outcomes:

After completion of the course the learner should be able to develop:

1. Handle specialized technology and update themselves with latest changes in technological world with ability to communicate effectively.
2. Be multi-skilled IT professional with good technical knowledge, management, leadership and entrepreneurship skills.
3. Be able to identify, formulate and model problems and find engineering solution based on a systems approach.

Course Contents:

Format of the Industrial Project Report	
1	INTRODUCTION
1.1	COMPANY PROFILE
1.2	EXISTING SYSTEM AND NEED OF SYSTEM
1.3	SCOPE OF WORK
1.4	OPERATING ENVIRONMENT – HARDWARE AND SOFTWARE
1.5	DESCRIPTION OF TECHNOLOGY USED
2	PROPOSED SYSTEM
2.1	PROPOSED SYSTEM

2.2	OBJECTIVE OF SYSTEM
2.3	ANALYSIS OF SYSTEM
2.4	FEASIBILITY STUDY
2.5	USER REQUIREMENTS
3	ANALYSIS AND DESIGN
3.1	OBJECT DIAGRAM
3.2	CLASS DIAGRAM
3.3	USE CASE DIAGRAM
3.4	SEQUENCE DIAGRAM
3.5	ACTIVITY DIAGRAM
3.6	MODULE HIERARCHY DIAGRAM
3.7	COMPONENT DIAGRAM
3.8	DEPLOYMENT DIAGRAM
3.9	MODULE SPECIFICATIONS
3.10	WEB SITE MAP DIAGRAM
3.11	USER INTERFACE DESIGN (SCREENS ETC.)
3.12	TABLE SPECIFICATIONS
3.13	TEST PROCEDURES AND IMPLEMENTATION
4	USER MANUAL
4.1	USER MANUAL
5	DRAWBACKS AND LIMITATIONS
6	PROPOSED ENHANCEMENTS
7	CONCLUSIONS
8	BIBLIOGRAPHY
9	ANNEXURE
	1. SAMPLE PROGRAM CODE
	2. OUTPUT REPORTS WITH DATA



Program Coordinator
MCA



Program Director/Asso. Dean
Academics
DoCSA, SoCSE



Dean
SoCSE



Registrar
MIT WPU